

Analysis Report

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1. Introduction

This report summarizes the insights from the following datasets:

- `fear_greed_index.csv` – Containing the sentiment score and value for 2644 trading days from 2018-02-01 to 2025-05-02.
- `historical_data.csv` - Containing the data for 211224 trades executed for 246 different coins from 32 different accounts.

The target is to evaluate the trader (represented by accounts in the `historical_data.csv` file) performance in different sentiment regimes, identify patterns from the same, and refine trading practices accordingly.

2. Assumptions

Since the explanations of the data specified in each of the columns were not available, some key assumptions had to be made for the analysis:

1. Fee column in `historical_data.csv` represents the transaction fee for each trade.
2. Closed PnL column `historical_data.csv` represents the dollar value of the P&L from each trade.

3. General Observations from Exploratory Data Analysis:

This section uncovers some rudimentary features of the data.

1. Markets are rarely neutral, and market sentiments tend to persist for some time before changing. Particularly noticeable is the persistence of fear in the bitcoin market sentiment at the onset of the Russia-Ukraine conflict in 2025 and the positive outlook that indicates the rally that took the bitcoin rally past USD 100000.

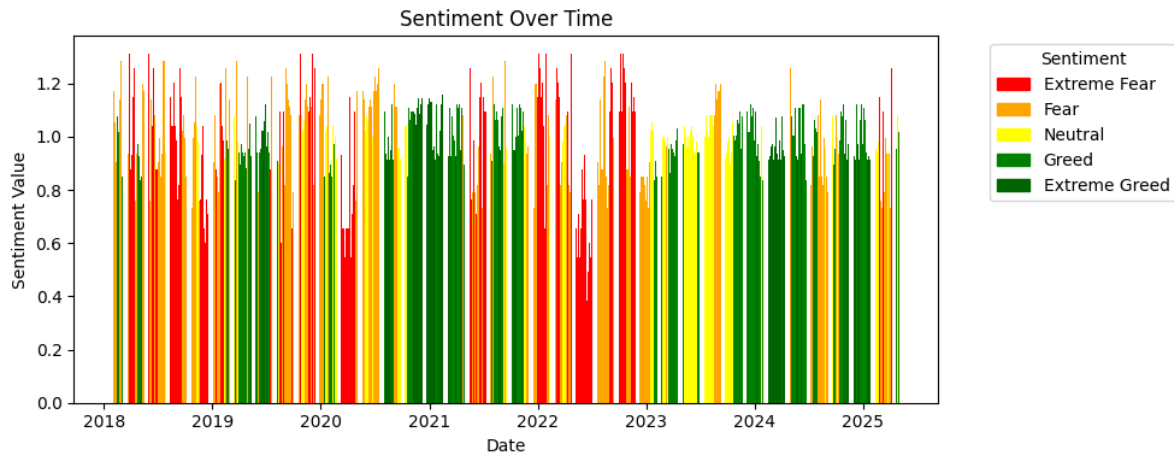


Figure-1

The above plot shows the values of the sentiment scores normalized with the corresponding average values of each sentiment for easier visualization.

On performing the Ad-Fuller test for stationarity, the p-value obtained is $1.1e-06$, showing that the series is stationary. The ACF plot validates the conclusion.

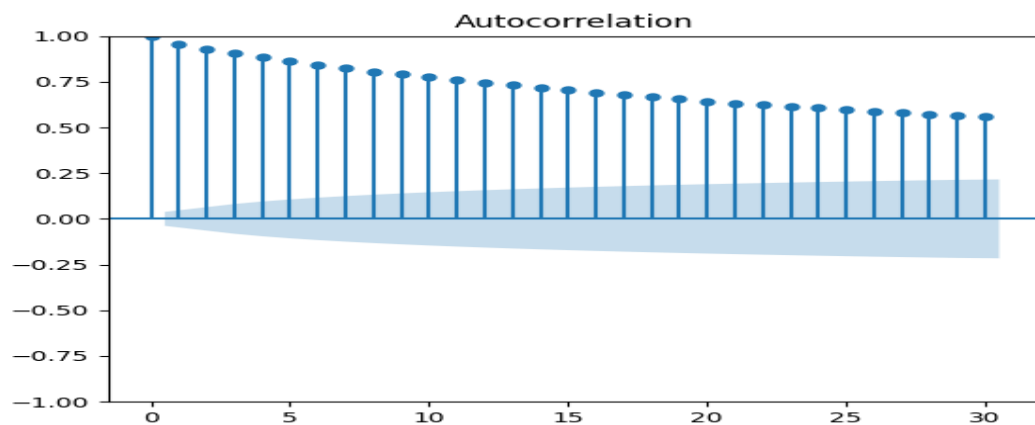


Figure-2

The p-value of 0 from the Ljung-Box Test and the value of 0.01599332388838119 obtained from the Durbin-Watson test indicate the presence of strong positive autocorrelation.

The Hurst exponent value of 0.5707973426247105 indicates the presence of trending properties, indicating moderate predictability using time series forecasting methods.

2. The Figure 3 shows the relative intensity of trading activities in different regimes as per market sentiment:

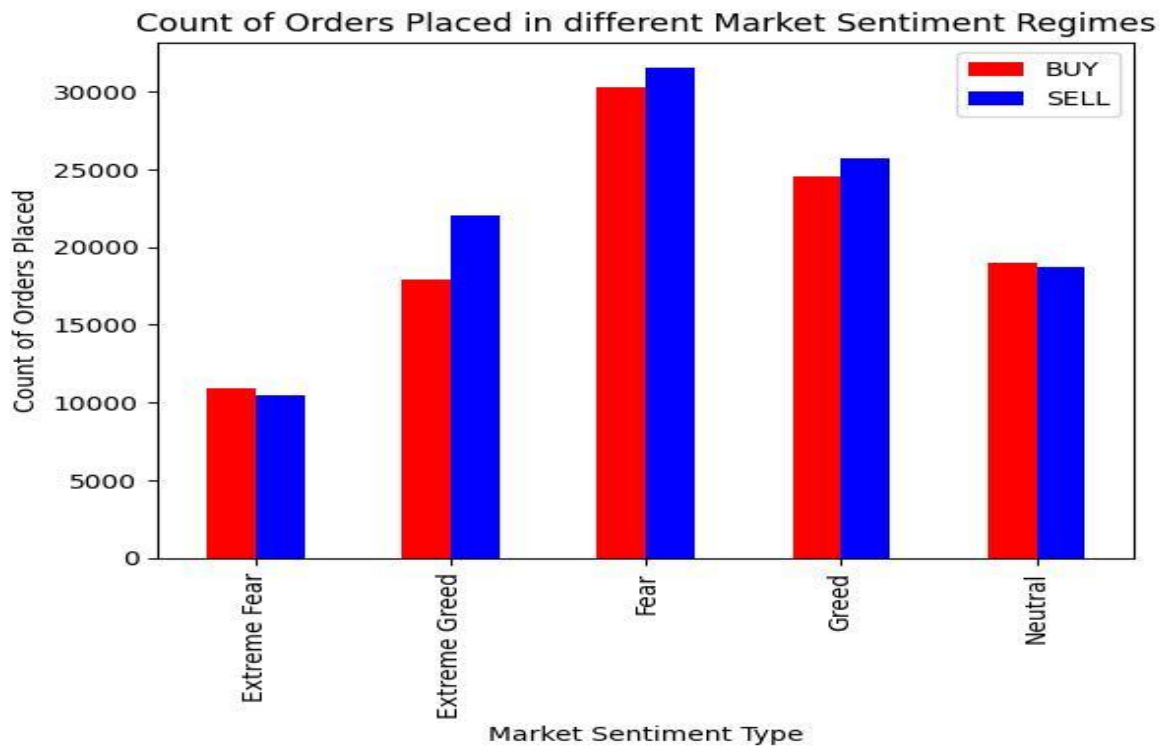


Figure-3

This indicates that the traders were most active on days when the general market sentiment was **fearful**.

- The total sum of PnL in different regimes of market sentiments indicates that the maximum was booked when the general market sentiment was **fearful**. However, this can be attributed to the general tendency of traders to square-off existing positions to protect the profits earned, which increased the total PnL. This can be supported by the plot comparing the transaction frequency in different market scenarios as in Figure-3. The average PnL was maximum when the general market sentiment was of **extreme greed**.

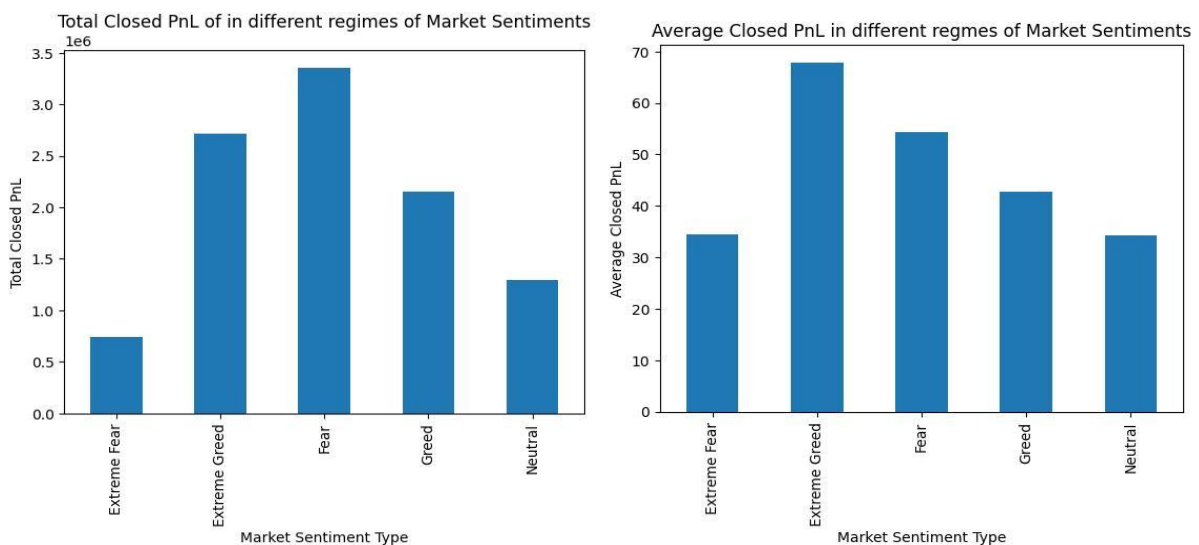


Figure-4

Figure-5

4. Considering only negative Closed PnLs, it can be observed that cumulatively, traders have lost the most when the market sentiment was that of **greed**. However, losses on individual transactions were recorded as maximum when the market was extremely fearful.

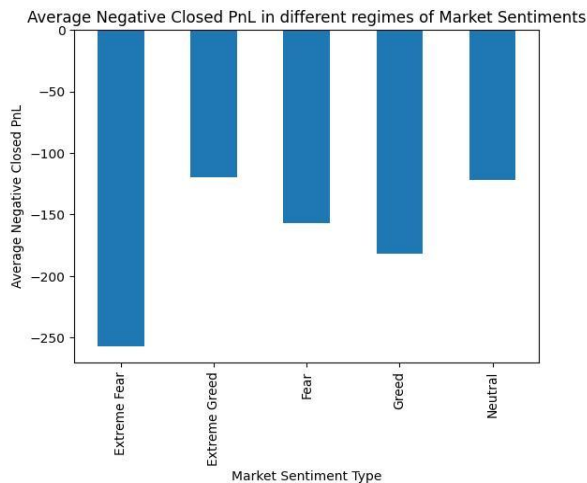


Figure-6

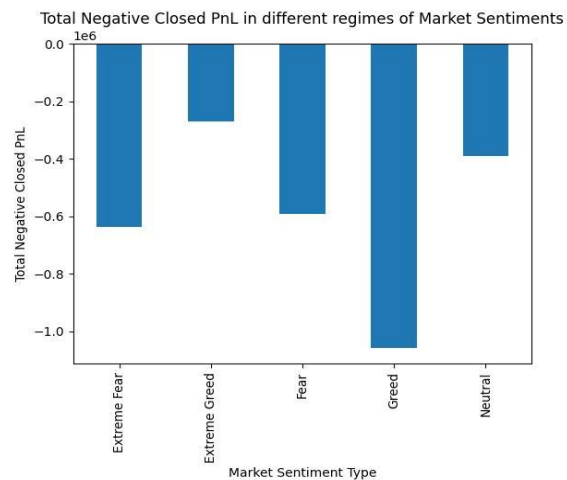


Figure-7

5. Considering only positive Closed PnLs, it can be observed that cumulatively, traders have registered the most cumulative gains when the market sentiment was that of **fear**, because of reasons discussed in point 3. However, considering gains on individual transactions fall in a short range for all market sentiments other than **neutral**, indicating the presence of a strong sentiment as a driver of gains and hence a suitable window for trading.

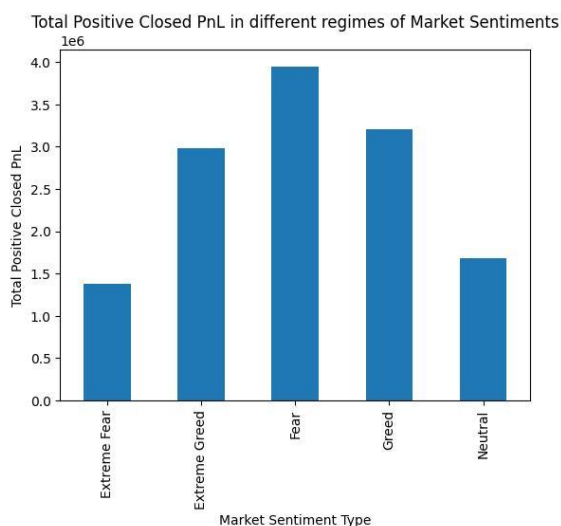


Figure-8

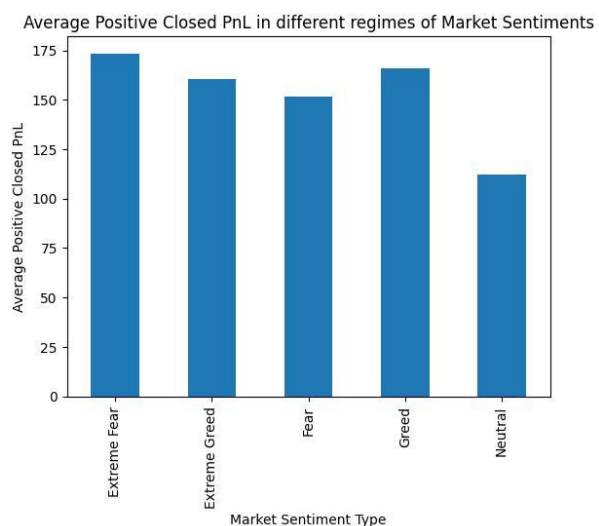


Figure-9

6. Considering the average positive and negative returns (figures 9 and 6, respectively), it can be noticed that in cases where market sentiments are **extremely fearful**, the

average losses significantly surpass the average gains. Figure 1 represents the frequency of stints when the markets enter such phases, which indicates that traders should be careful when trading in these times.

- The following plot shows the overall distribution of the nature of transactions. This reflects that traders are more comfortable in taking short positions when the general market sentiment is that of **greed**.

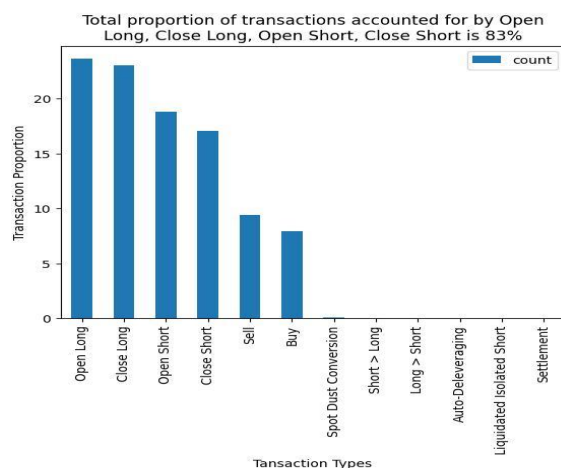


Figure-8a

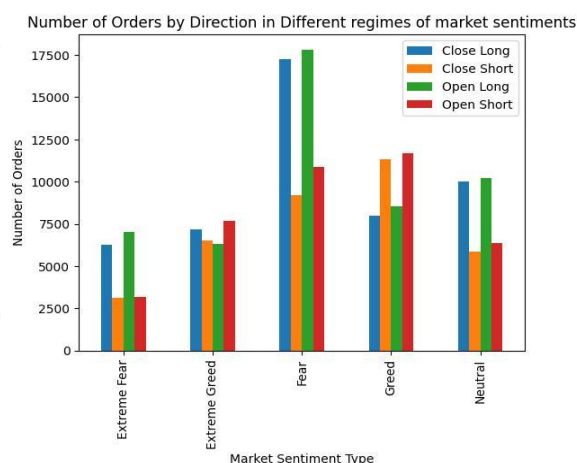


Figure-9a

- The Figure 9b show that on short positions have remained more profitable on average than long positions. Figure 8a shows that the average trade sizes associated with long positions are higher.

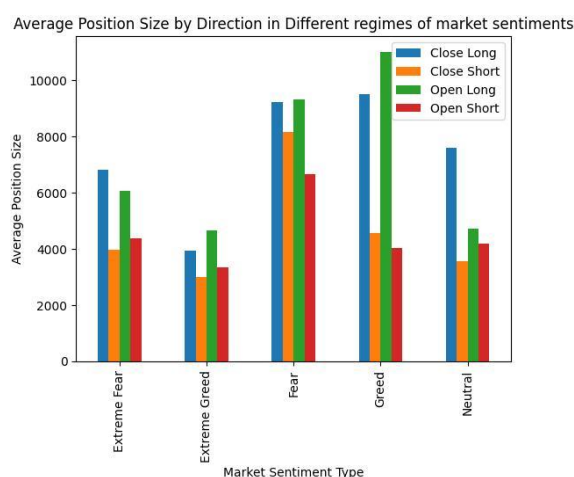


Figure-8b

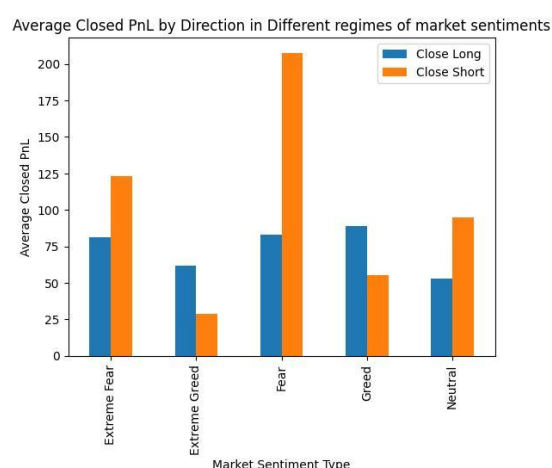


Figure-9b

- The similarity between the plots for average fees and average trade sizes justifies the direct relation between transaction fees and transaction demands on blockchain, indicating that trading during **fearful** markets can be significantly costlier.

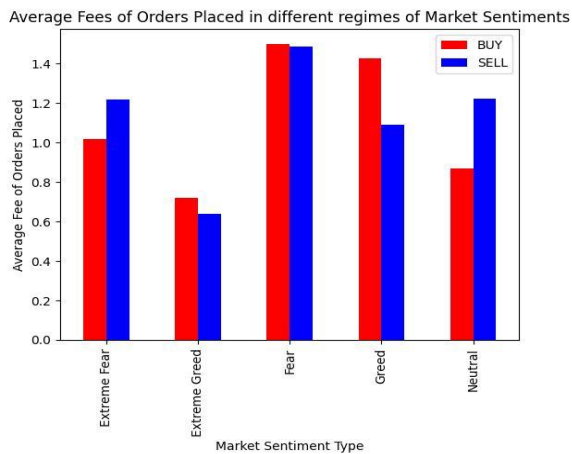


Figure-10

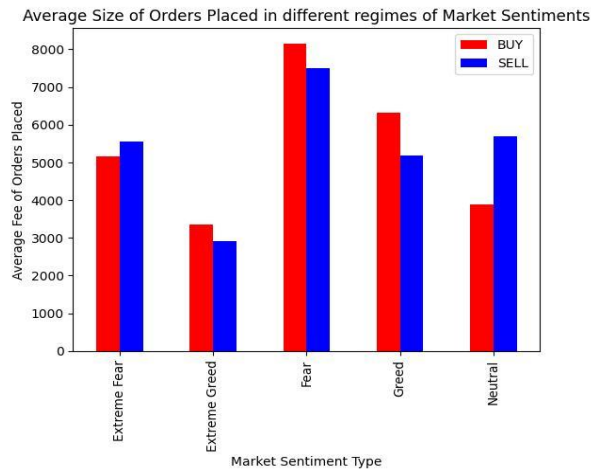


Figure-11

10. Despite a total of 246 different coins being traded, more than 75% transactions overall, categorically “BUY” and categorically “SELL” were accounted for by the top 10 (4% of total) coins traded, as shown in the plots below.

Total Contribution of Top 10 Traded Coins in Total Number of Trades = 76.57%

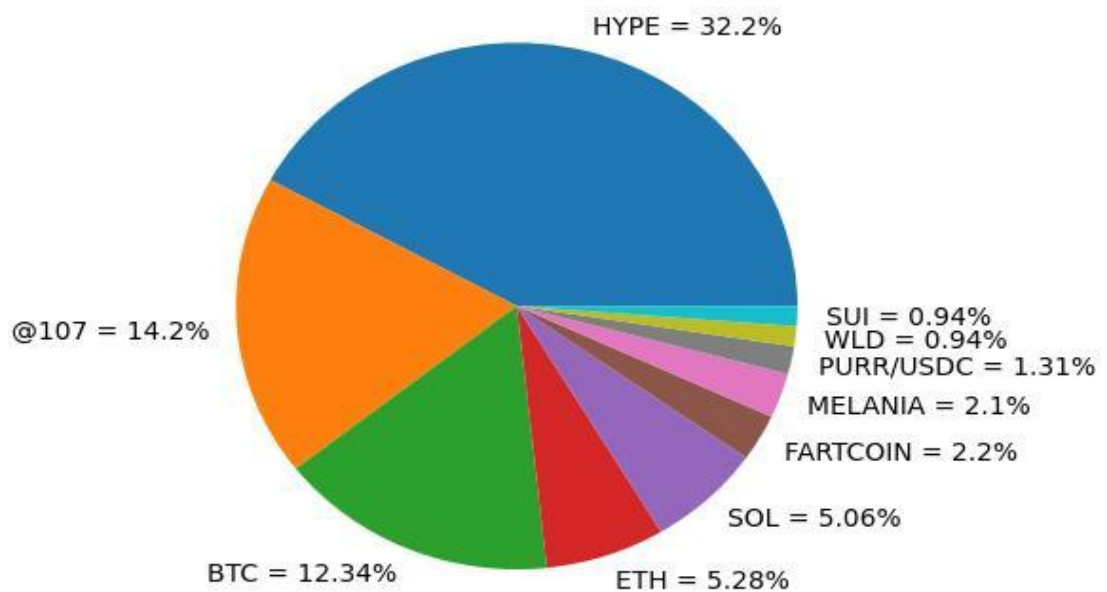


Figure-12

Total Contribution of Top 10 Traded Coins in Total Number of Buy Trades = 77.24%

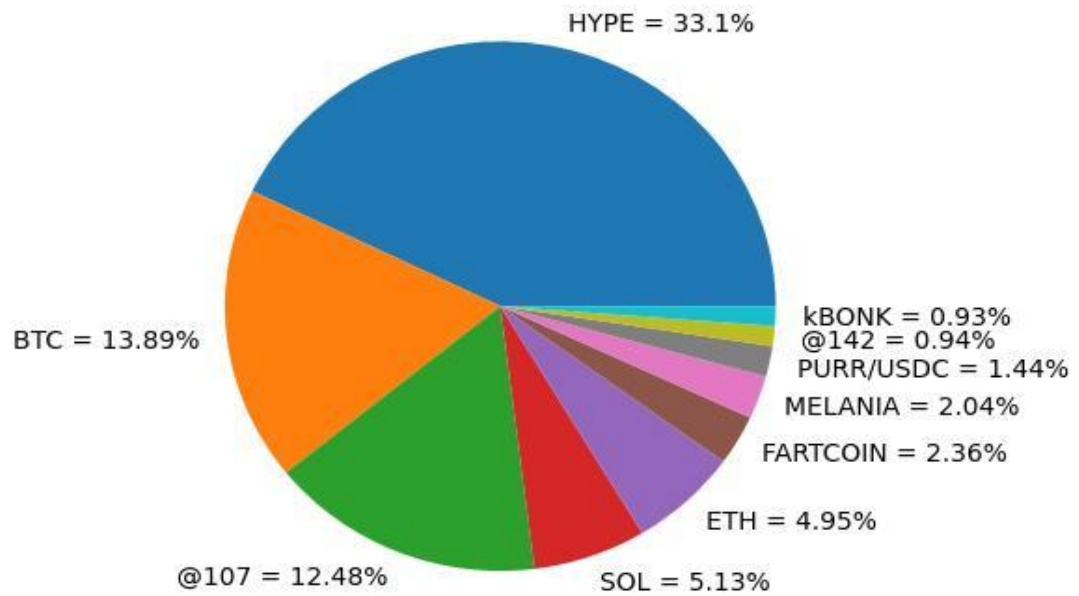


Figure-13

Total Contribution of Top 10 Traded Coins in Total Number of Sell Trades = 76.02%

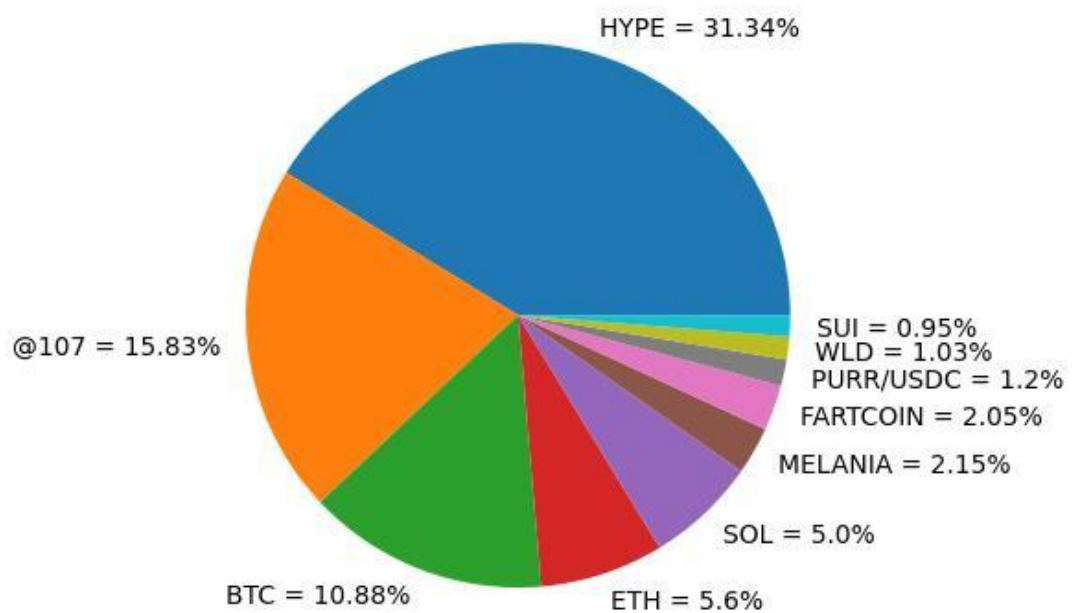


Figure-14

This indicates that, in considering coin-specific performance, considering the top 10 coins will be sufficient. HYPE, BTC, ETH, SOL, @107, FARTCOIN, and MELANIA are common in all three sets. The following figures show the contribution of top 10 coins by the total number of trades in each market regime.

Total Contribution of Top 10 Traded Coins in Total Number of Trades = 67.19% in Extreme Greed Regime

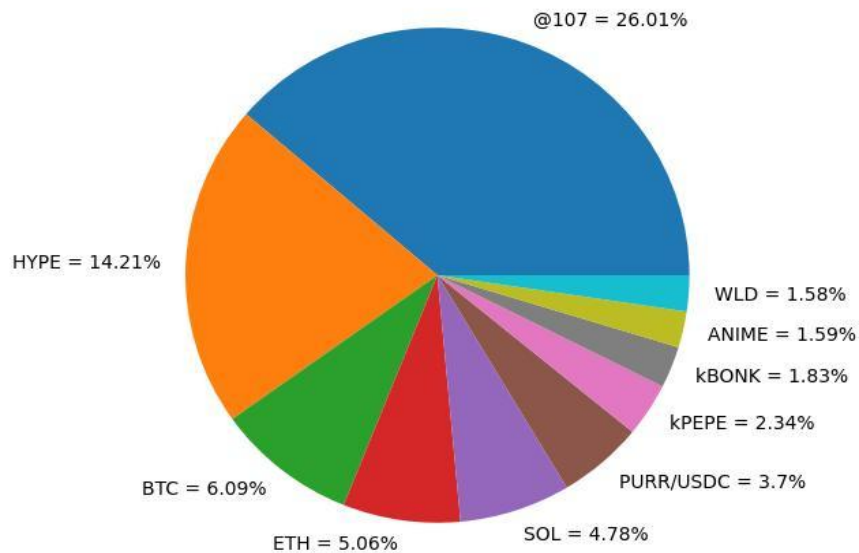


Figure-15

Total Contribution of Top 10 Traded Coins in Total Number of Trades = 67.82% in Greed Regime

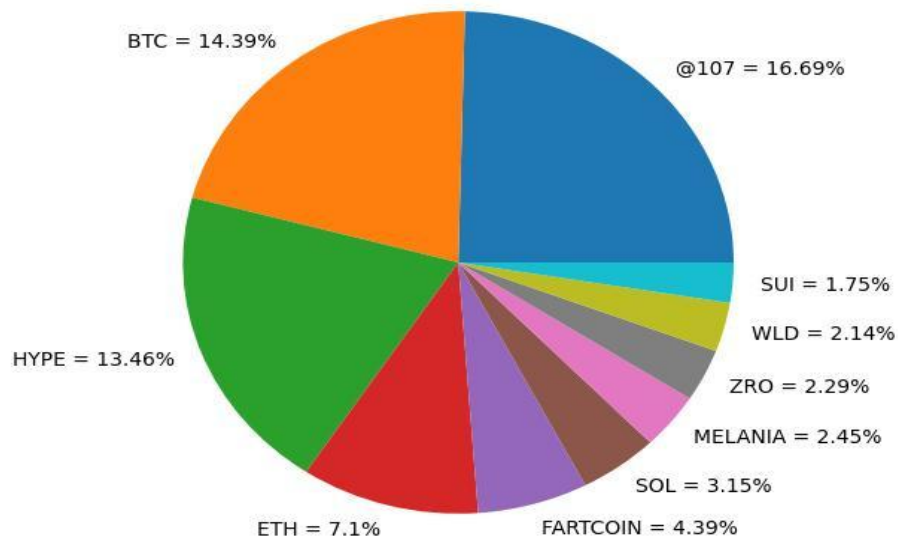


Figure-16

Total Contribution of Top 10 Traded Coins in Total Number of Trades = 83.62% in Neutral Regime

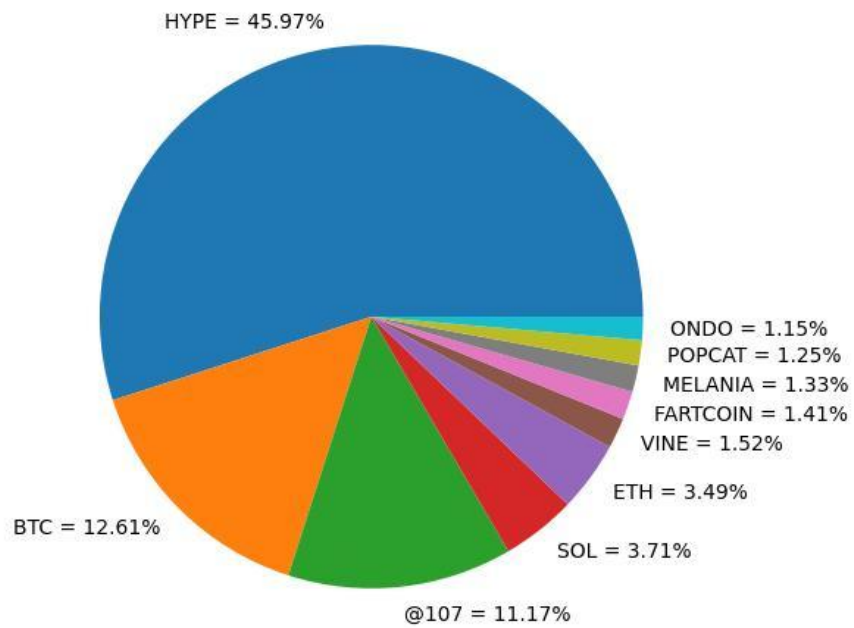


Figure-17

Total Contribution of Top 10 Traded Coins in Total Number of Trades = 88.12% in Fear Regime

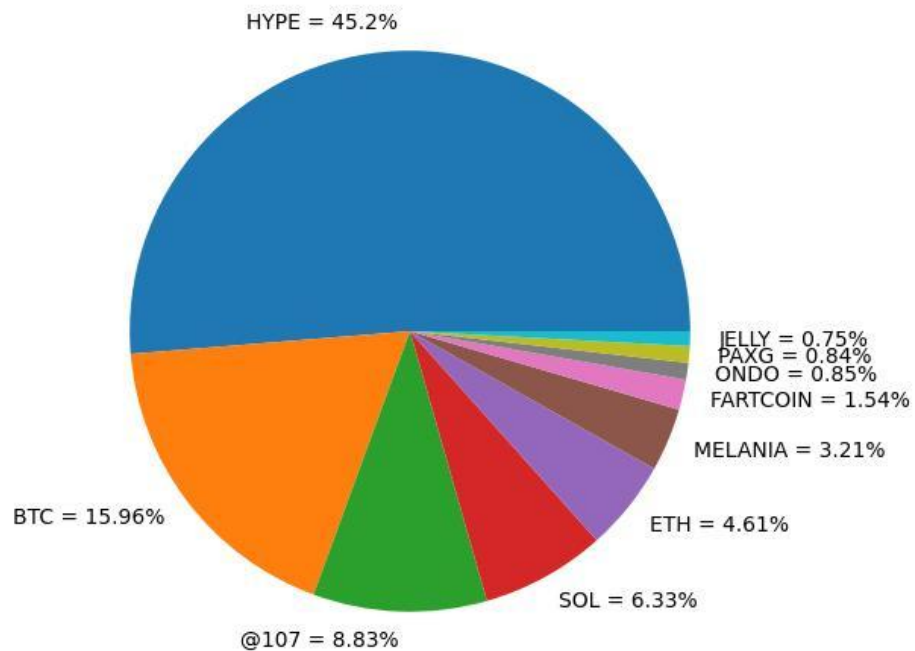


Figure-18

Total Contribution of Top 10 Traded Coins in Total Number of Trades = 90.72% in Extreme Fear Regime

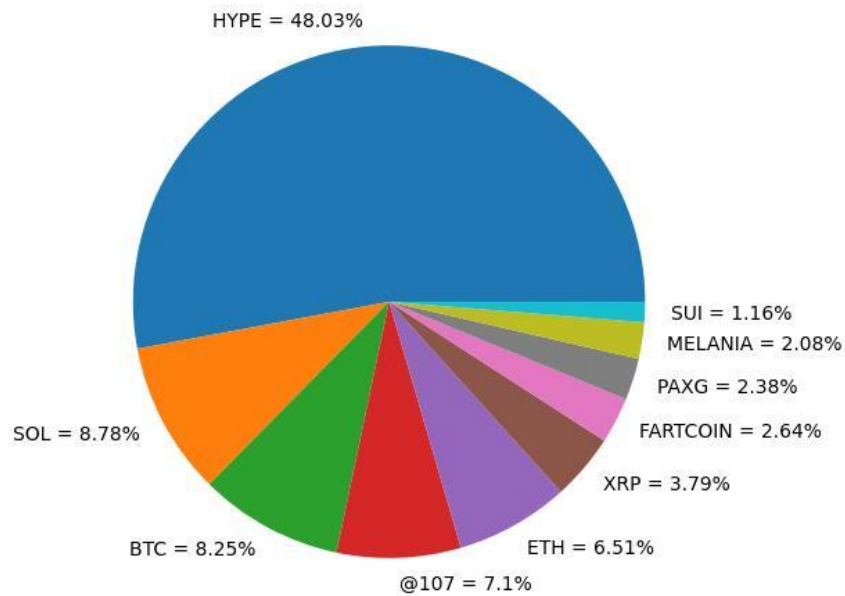


Figure-19

The figures clearly reflect the presence of the top 10 coins by overall trades in each of the market regimes. Two interesting observations can be made in this regard:

- The increase in the trading of HYPE coin as market sentiments move from extremely greedy to extremely fearful.
- The increase in concentration of transactions in selected coins as market sentiments move from extremely greedy to extremely fearful.

11. The skewness in the transactions overall and during different sentiment regimes show that considering average Closed PnL per transaction is a better indicator for identifying the relative performance of the coins from the trade data. The following plots show the top-10 coins with the highest average closed PnL in each sentiment regime.

Comparison of Top 10 Coins with highest Mean Booked PnL Neutral Regime

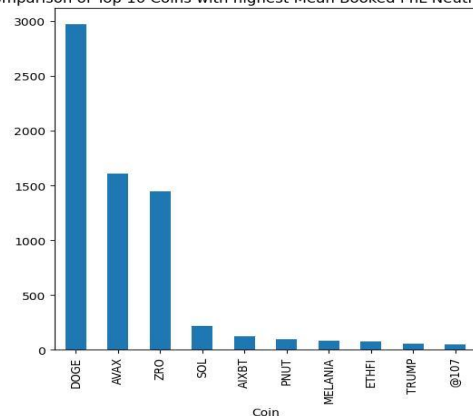


Figure-20

Comparison of Top 10 Coins with highest Mean Booked PnL Greed Regime

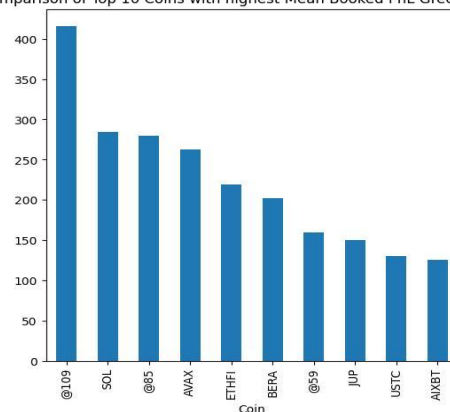


Figure-21

Comparison of Top 10 Coins with highest Mean Booked PnL Fear Regime

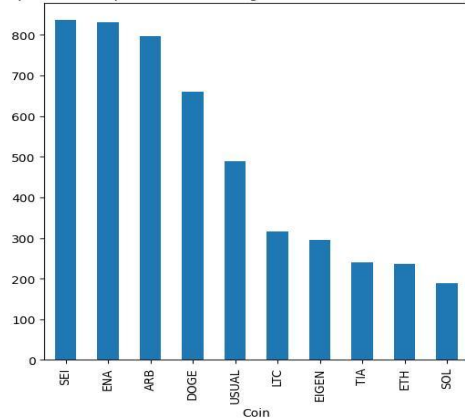


Figure-22

Comparison of Top 10 Coins with highest Mean Booked PnL Extreme Greed Regime

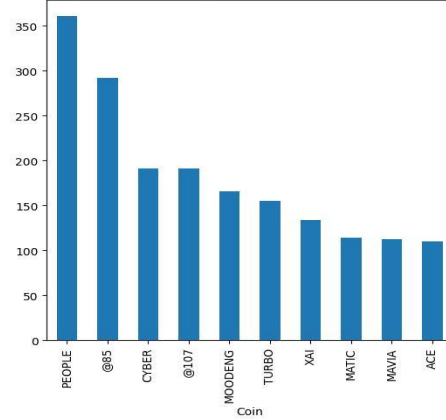


Figure-23

Comparison of Top 10 Coins with highest Mean Booked PnL Extreme Fear Regime

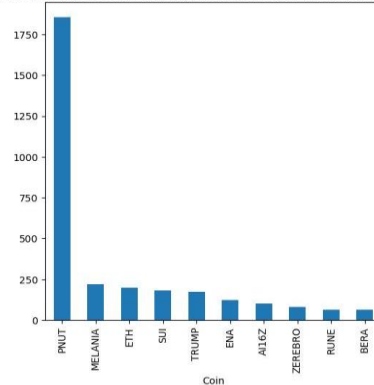


Figure-24

4. Trader Analytics

To analyse trader performance and hence understand the influence of BTC market sentiment on their trading activities, the following method was followed:

1. Each account was mapped to a representative name for easier representation in charts and data tables. The mapping is as follows:

Account Address	Name
0xbee1707d6b44d4d52bfe19e41f8a828645437aab	Trader 1
0xbaaaf6571ab7d571043ff1e313a9609a10637864	Trader 2
0xa0feb3725a9335f49874d7cd8eaad6be45b27416	Trader 3
0x8477e447846c758f5a675856001ea72298fd9cb5	Trader 4
0xb1231a4a2dd02f2276fa3c5e2a2f3436e6bfed23	Trader 5
0x28736f43f1e871e6aa8b1148d38d4994275d72c4	Trader 6
0x513b8629fe877bb581bf244e326a047b249c4ff1	Trader 7
0x75f7eeb85dc639d5e99c78f95393aa9a5f1170d4	Trader 8
0x47add9a56df66b524d5e2c1993a43cde53b6ed85	Trader 9
0x4f93fead39b70a1824f981a54d4e55b278e9f760	Trader 10
0x23e7a7f8d14b550961925fbfdaa92f5d195ba5bd	Trader 11

0xb899e522b5715391ae1d4f137653e7906c5e2115	Trader 12
0x8170715b3b381dff7062c0298972d4727a0a63b	Trader 13
0x4acb90e786d897ecffb614dc822eb231b4ffb9f4	Trader 14
0x083384f897ee0f19899168e3b1bec365f52a9012	Trader 15
0x271b280974205ca63b716753467d5a371de622ab	Trader 16
0x39cef799f8b69da1995852eea189df24eb5cae3c	Trader 17
0x2c229d22b100a7beb69122eed721cee9b24011dd	Trader 18
0x92f17e8d81a944691c10e753af1b1baae1a2cd0d	Trader 19
0xbd5fead7180a9c139fa51a103cb6a2ce86ddb5c3	Trader 20
0x8381e6d82f1affd39a336e143e081ef7620a3b7f	Trader 21
0x72743ae2822edd658c0c50608fd7c5c501b2afbd	Trader 22
0x7f4f299f74eec87806a830e3caa9afa5f2b9db8f	Trader 23
0x72c6a4624e1dffa724e6d00d64ceae698af892a0	Trader 24
0x430f09841d65beb3f27765503d0f850b8bce7713	Trader 25
0x6d6a4b953f202f8df5bed40692e7fd865318264a	Trader 26
0x3998f134d6aaa2b6a5f723806d00fd2bbbbce891	Trader 27
0xae5eacaf9c6b9111fd53034a602c192a04e082ed	Trader 28
0xaf40fdc468c30116bd3307bcbf4a451a7ebf1deb	Trader 29
0xa520ded057a32086c40e7dd6ed4eb8efb82c00e0	Trader 30
0x420ab45e0bd8863569a5efbb9c05d91f40624641	Trader 31
0x3f9a0aad7f04a7c9d75dc1b5a6ddd6e36486cf6	Trader 32

2. The performance of traders was evaluated, taking 5 factors into account:

- **Sum Closed PnL:** Total booked P&L by trader. This evaluates the cumulative performance of the trader over the period.
- **Mean Closed PnL:** Average P&L booked by a trader. This evaluates the precision of the profit motive of the trader while executing trades during the period. Since Closed PnL was recorded only for close positions, the evaluated factor was adjusted accordingly to represent the intended metric with reasonable accuracy.
- **Sum Fee:** Total transaction fees paid by a trader over the period. This evaluates the total transaction volume as well as the additional deduction to the registered P&L. For specific cases, like Trader 12, if the transaction fees are added to the total closed P&L, which is itself negative, then the actual losses would be 25% more than what is reflected as only from trading.
- **Mean Fee:** Average transaction fees paid by a trader. This metric is indicative of the prevailing BTC market sentiment when a trader was most active, and his or her order sizes, as the clear relation between transaction fees and market sentiment is evident from point 9 of the general observations.
- **Count Transaction Outcome:** Count of total trades for a trader. This metric evaluates the relative activity of a trader in the said period

- **Mean Transaction Outcome:** Proportion of winning trades for a trader. Calculated by the average of the trades where P&L recorded was positive, this metric evaluates the accuracy of the profit motive of the trader while executing trades during the period. Since Closed PnL was recorded only for close positions, the evaluated factor was adjusted accordingly to represent the intended metric with reasonable accuracy.

The traders were ranked in these 5 categories according to their overall performance as well as performance during individual regimes.

4.1. Evaluation metrics for the entire period

4.1.1. Trader Performance

Account	Sum Closed Pnl	Mean Closed Pnl	Sum Fee	Mean Fee	Mean Transaction Outcome	Count Transaction Outcome
Trader 1	761515.53	33.44	8167.01	0.36	57%	22771.00
Trader 10	203533.85	59.65	20383.79	5.97	59%	3412.00
Trader 11	44748.77	10.59	1696.60	0.40	64%	4225.00
Trader 12	30309.08	12.27	14216.30	5.76	52%	2470.00
Trader 13	10559.87	4.51	551.06	0.24	14%	2339.00
Trader 14	649309.77	289.74	4354.93	1.94	87%	2241.00
Trader 15	-24781.01	-11.76	3008.88	1.43	9%	2107.00
Trader 16	17705.06	7.89	4485.04	2.00	5%	2243.00
Trader 17	-7021.44	-3.62	735.30	0.38	16%	1941.00
Trader 18	168691.45	81.89	1393.85	0.68	82%	2060.00
Trader 19	88161.05	93.00	1158.48	1.22	45%	948.00
Trader 2	940163.81	94.04	5415.25	0.54	99%	9997.00
Trader 20	7706.61	5.32	2250.57	1.55	23%	1448.00
Trader 21	63031.47	83.37	1780.63	2.36	73%	756.00
Trader 22	40209.75	40.53	638.58	0.64	40%	992.00
Trader 23	9245.96	10.65	423.66	0.49	28%	868.00
Trader 24	-18216.40	-23.03	164.88	0.21	2%	791.00
Trader 25	416541.87	445.50	328.35	0.35	64%	935.00
Trader 26	106196.49	173.52	136.98	0.22	54%	612.00
Trader 27	-40697.99	-64.29	73.03	0.12	49%	633.00
Trader 28	67845.62	293.70	225.03	0.97	100%	231.00
Trader 29	21758.83	94.60	359.59	1.56	75%	230.00
Trader 3	103244.02	18.36	1474.13	0.26	93%	5622.00
Trader 30	72846.48	279.11	88.46	0.34	92%	261.00
Trader 31	43.37	0.15	174.78	0.62	4%	282.00
Trader 32	53496.25	330.22	113.19	0.70	98%	162.00
Trader 4	30740.57	3.83	2107.22	0.26	7%	8034.00
Trader 5	1486783.66	185.15	7346.66	0.91	31%	8030.00
Trader 6	125675.82	19.01	1099.89	0.17	58%	6612.00
Trader 7	888477.92	151.41	38510.90	6.56	71%	5868.00

Trader 8	219089.39	44.14	1371.67	0.28	80%	4964.00
Trader 9	52232.15	11.76	591.15	0.13	26%	4443.00

4.1.2. Trader Rank

Account	Sum Closed Pnl rank	Mean Closed Pnl rank	Sum Fee rank	Mean Fee rank	Mean Transaction Outcome rank	Count Transaction Outcome rank
Trader 1	4	17	4	21	16	1
Trader 10	8	14	2	2	14	11
Trader 11	19	23	13	19	13	10
Trader 12	22	20	3	3	18	12
Trader 13	25	26	22	27	27	13
Trader 14	5	4	8	6	6	15
Trader 15	31	30	9	9	28	16
Trader 16	24	24	7	5	30	14
Trader 17	29	29	19	20	26	18
Trader 18	9	13	15	14	7	17
Trader 19	13	11	17	10	20	21
Trader 2	2	10	6	17	2	2
Trader 20	27	25	10	8	25	19
Trader 21	16	12	12	4	10	25
Trader 22	20	16	20	15	21	20
Trader 23	26	22	23	18	23	23
Trader 24	30	31	28	29	32	24
Trader 25	6	1	25	22	12	22
Trader 26	11	7	29	28	17	27
Trader 27	32	32	32	32	19	26
Trader 28	15	3	26	11	1	30
Trader 29	23	9	24	7	9	31
Trader 3	12	19	14	26	4	7
Trader 30	14	5	31	23	5	29
Trader 31	28	28	27	16	31	28
Trader 32	17	2	30	13	3	32
Trader 4	21	27	11	25	29	3
Trader 5	1	6	5	12	22	4
Trader 6	10	18	18	30	15	5
Trader 7	3	8	1	1	11	6
Trader 8	7	15	16	24	8	8
Trader 9	18	21	21	31	24	9

4.1.3. Analysis

As evident from the highlighted rows above, each metric focuses on a unique feature of a trader's behaviour. Focusing on the top rankers in each category helps us identify the behavioural patterns and the influence of selected factors on the other metrics.

1. Trader 1: Highest number of transactions overall. The relatively lower rank in mean transaction fees indicate effective position sizing techniques that drove the overall performance.
2. Trader-5: Although the cumulative P&L is the highest, the win percentage is only 31%.
3. Trader 7: The highest in total and average fees paid. The relatively higher rank for total transaction and the overall P&L rank alongside the win percentage, indicates a well rounded trading approach.
4. Trader-17: Recorded the lowest activity overall.
5. Trader 25: Recorded the highest mean PnLs overall, however the ranks in the fees as well as the win percentage of 64% indicate reliance on opportune trading strategies with relatively large positions.
6. Trader 24: Recorded the worst performance in all trading matrices other than trade count and win percentage.
7. Trader 26: Recorded the worst win percentage.
8. Trader 28: Recorded the highest win percentage but transaction count indicates relatively fewer transactions made. The mean PnL rank however indicates opportune trading techniques.

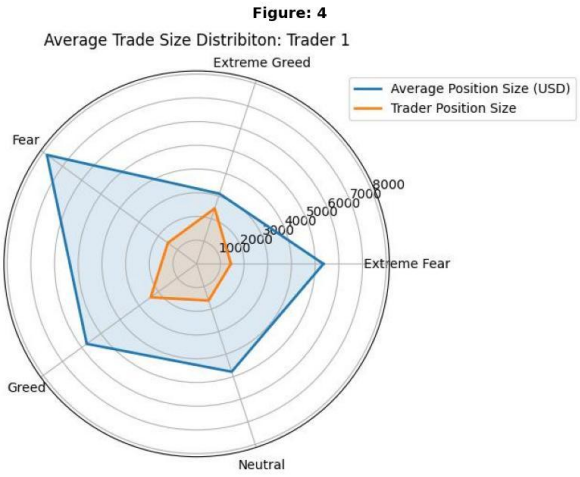
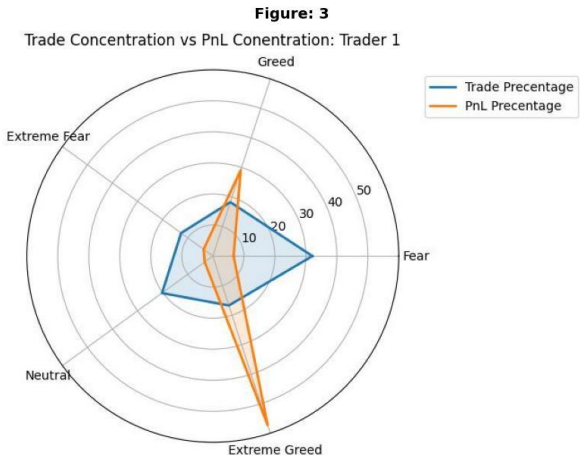
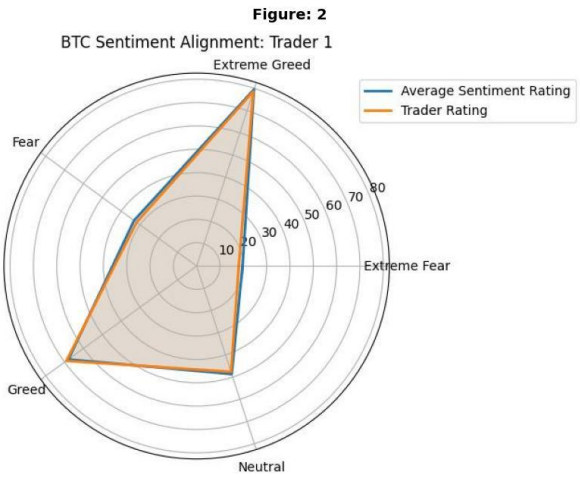
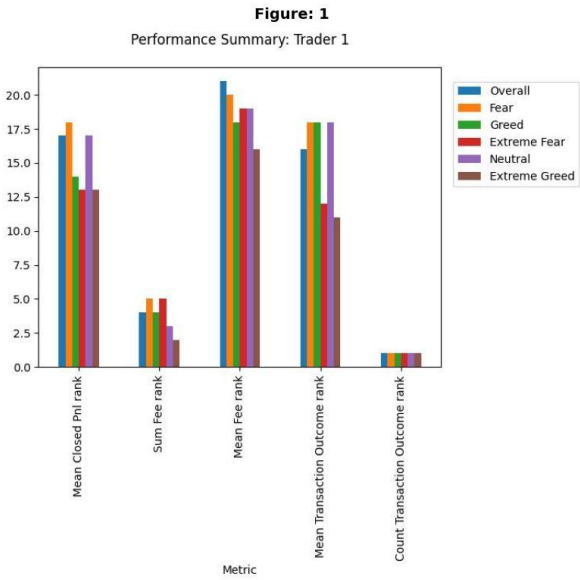
4.2. Individual Performance Ranks

The following plots summarize the relative performance of individual traders overall and in different regimes of market sentiment by plotting their ranks in the same settings to evaluate their behavioral and performance metrics.

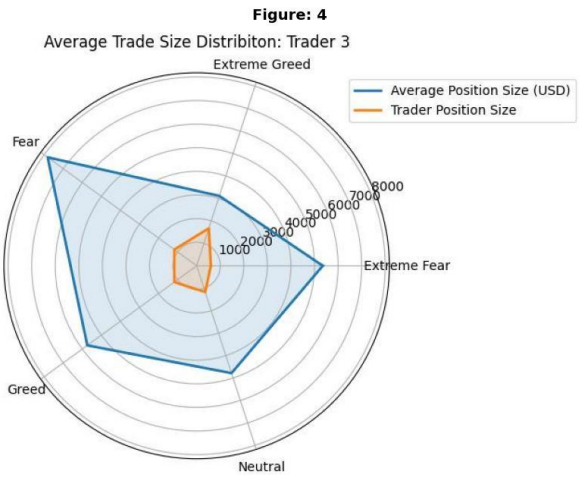
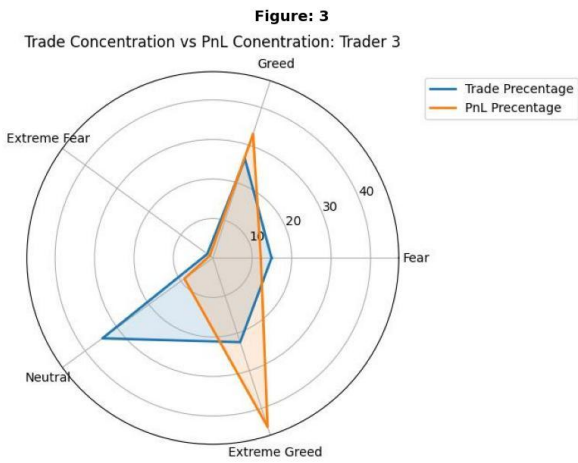
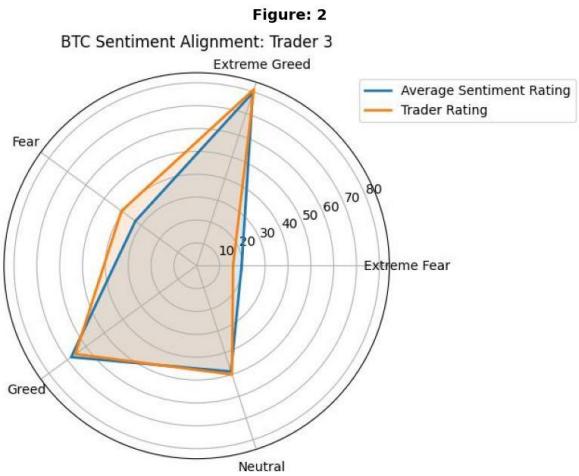
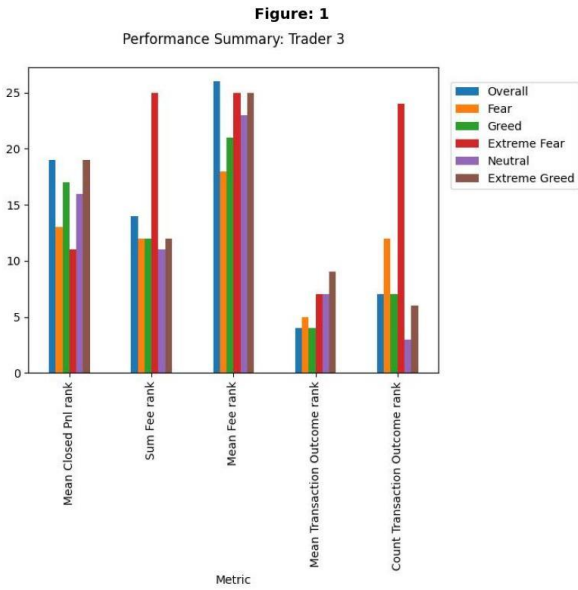
For each trader, a set of 4 plots has been furnished, representing the following:

1. **Figure 1**: Rank of the trader in different metrics overall and in different regimes of market sentiments.
2. **Figure 2**: Comparison of the average rating of sentiment score for each sentiment regime for all trades with the average of sentiment score on the days the traders executed their trades for each sentiment.
3. **Figure 3**: Distribution of the proportion of trades and that of total PnL for each trader in each regime of market sentiment.
4. **Figure 3**: Comparison of the average position size (in USD) of trades executed by traders in each regime of market sentiment with that of the trader.

Combined BTC Market Sentiment Analysis for Trader 1
Account: 0xbee1707d6b44d4d52bfe19e41f8a828645437aab

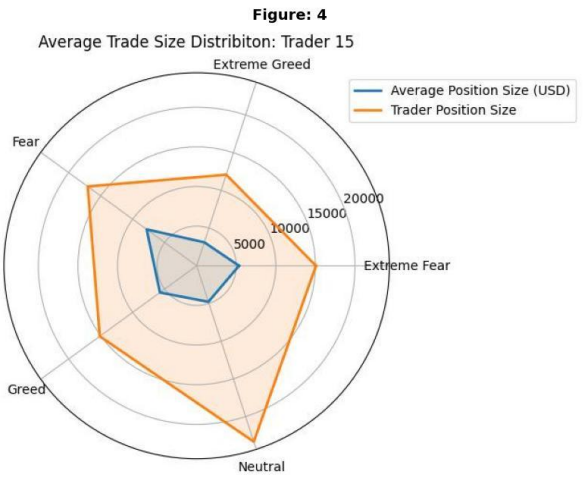
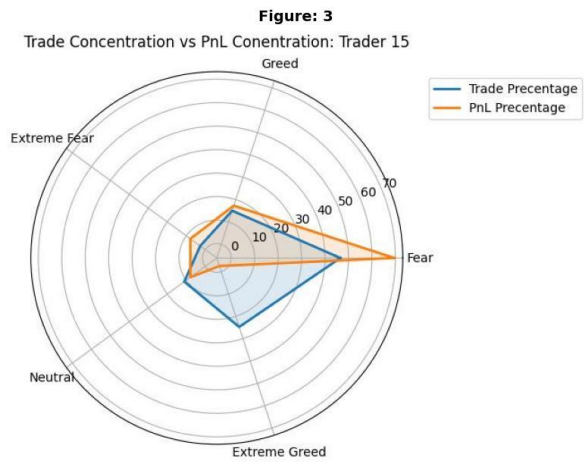
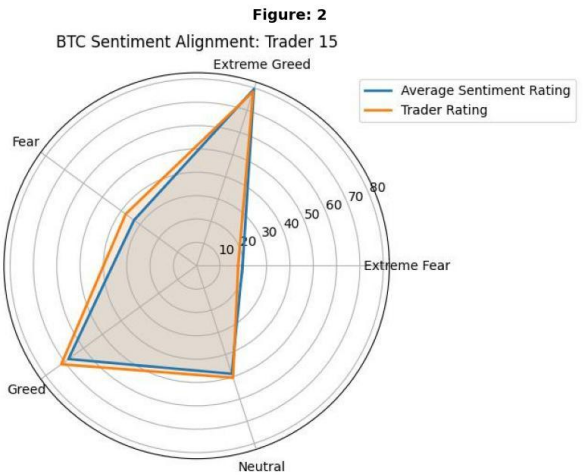
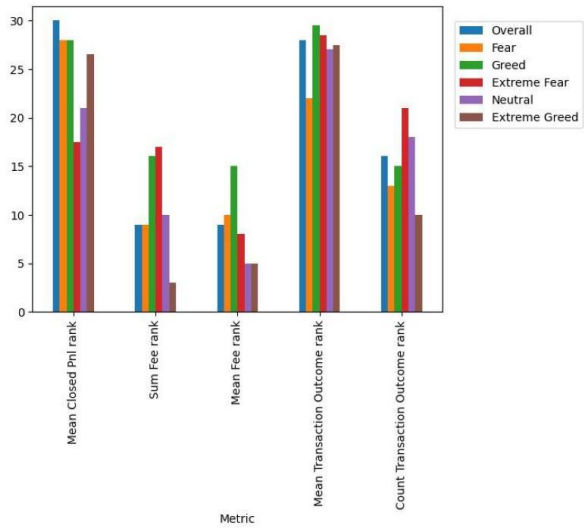


Combined BTC Market Sentiment Analysis for Trader 3
Account: 0xa0feb3725a9335f49874d7cd8ead6be45b27416

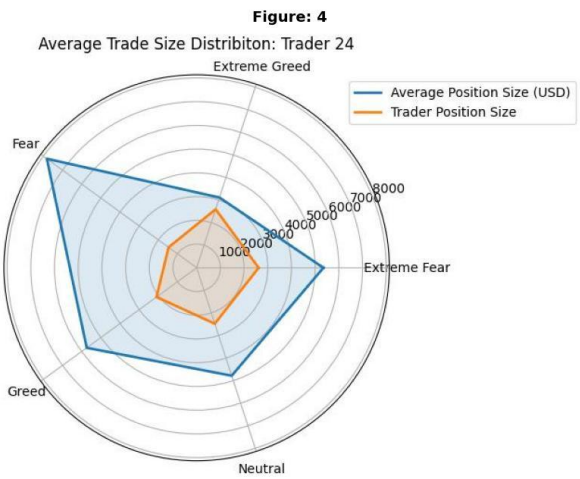
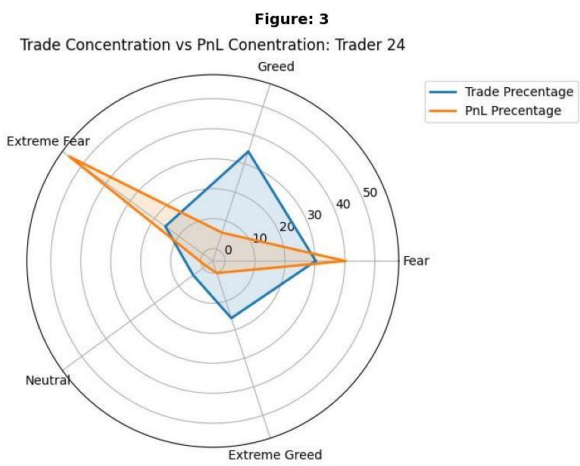
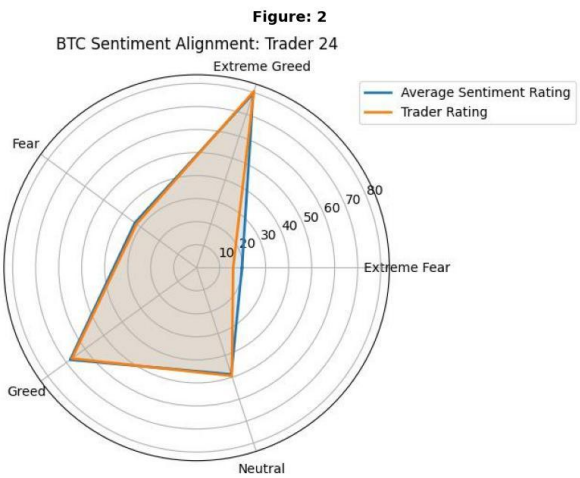
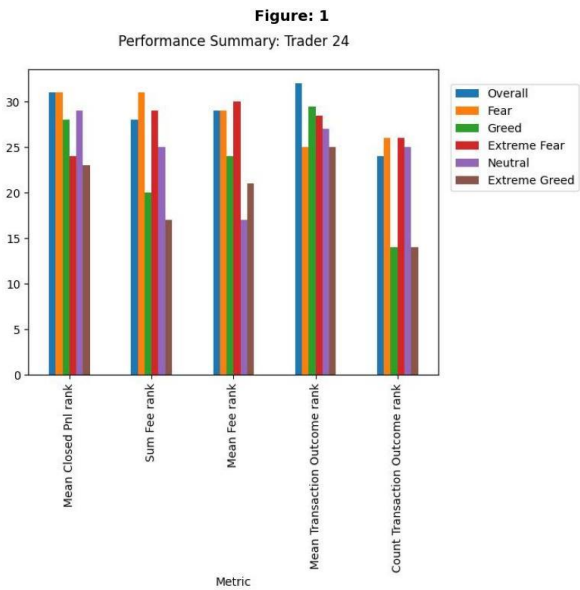


Combined BTC Market Sentiment Analysis for Trader 15
Account: 0x083384f897ee0f19899168e3b1bec365f52a9012

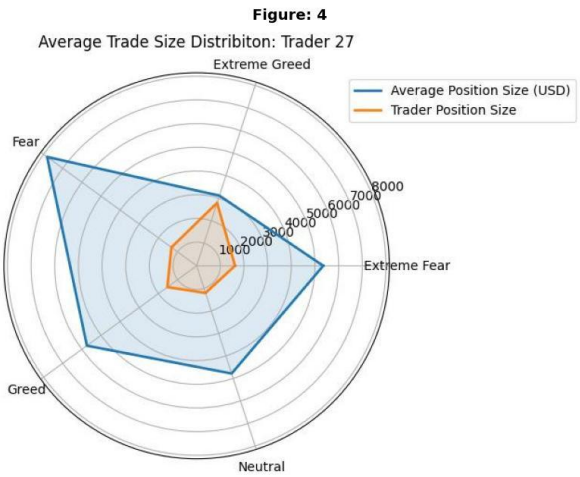
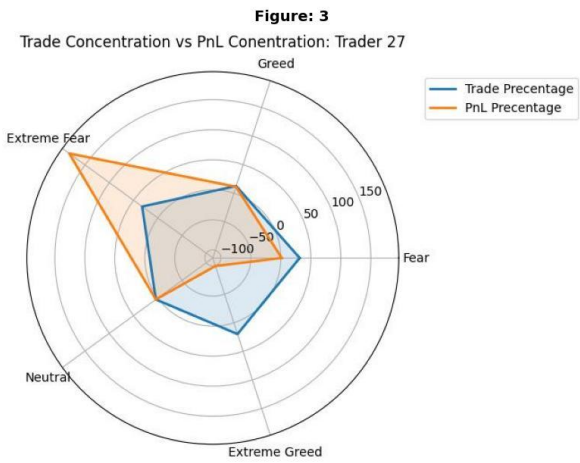
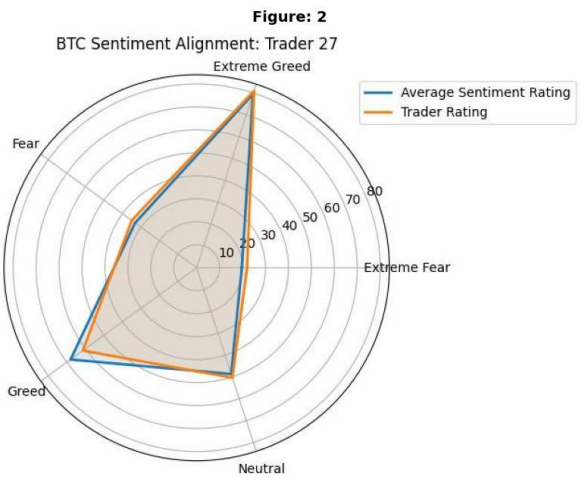
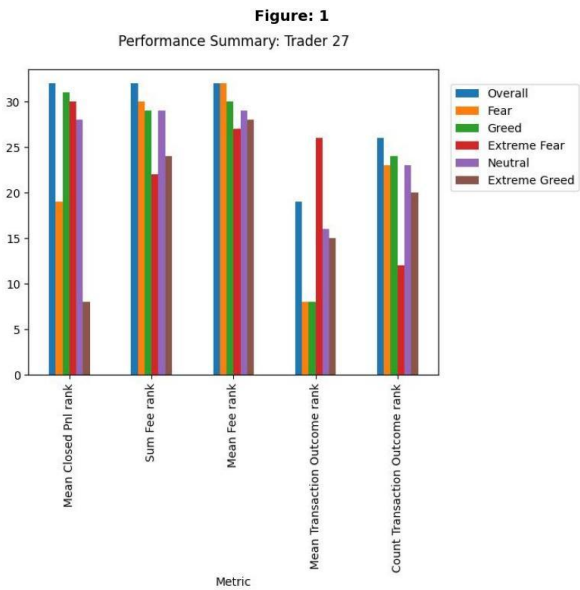
Figure: 1
Performance Summary: Trader 15



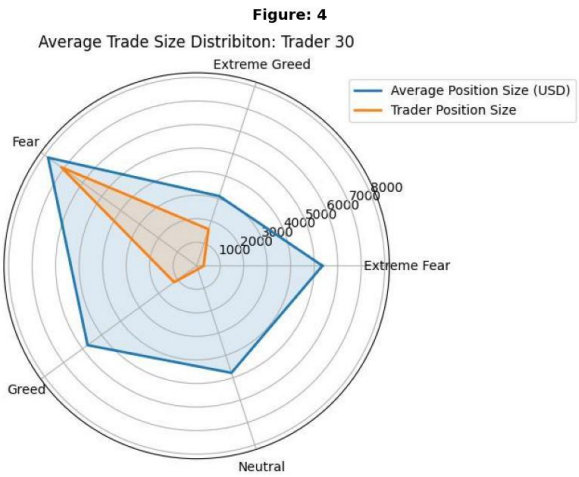
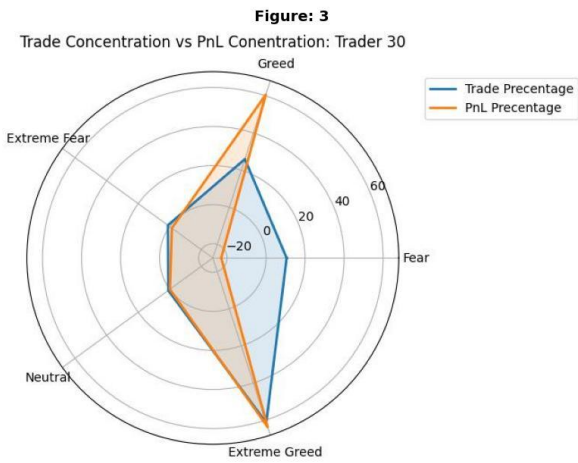
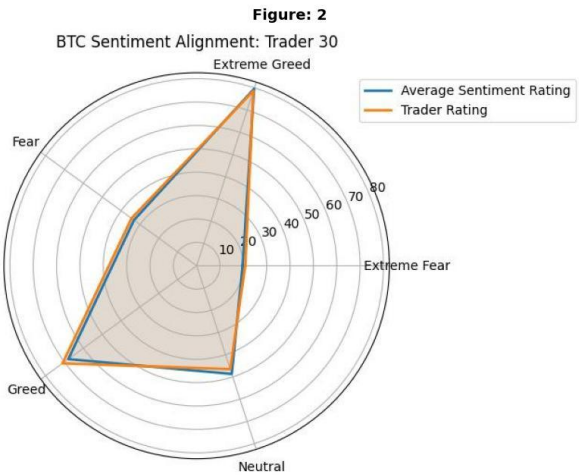
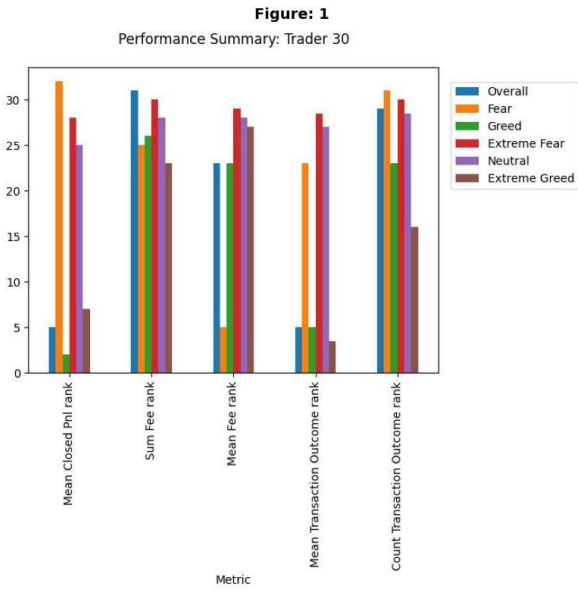
Combined BTC Market Sentiment Analysis for Trader 24
Account: 0x72c6a4624e1dffa724e6d00d64ceae698af892a0



Combined BTC Market Sentiment Analysis for Trader 27
Account: 0x3998f134d6aaa2b6a5f723806d00fd2bbbbce891



Combined BTC Market Sentiment Analysis for Trader 30
Account: 0xa520ded057a32086c40e7dd6ed4eb8efb82c00e0



Combined BTC Market Sentiment Analysis for Trader 8
Account: 0x75f7eeb85dc639d5e99c78f95393aa9a5f1170d4

Figure 1
Performance Summary: Trader 8

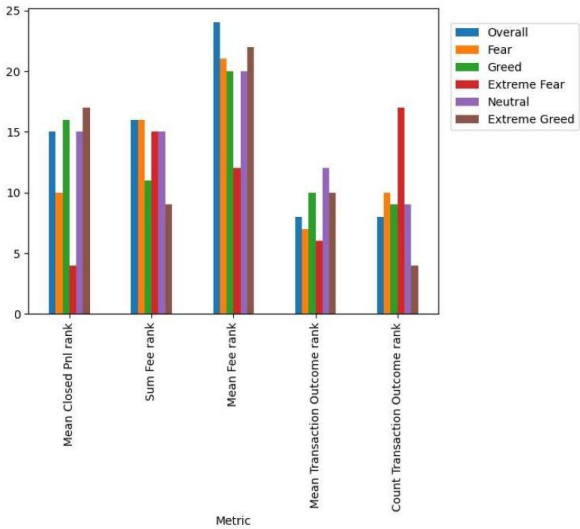


Figure 2
BTC Sentiment Alignment: Trader 8

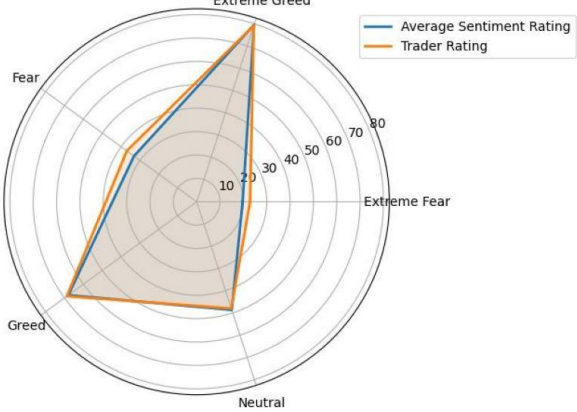


Figure 3
Trade Concentration vs PnL Concentration: Trader 8

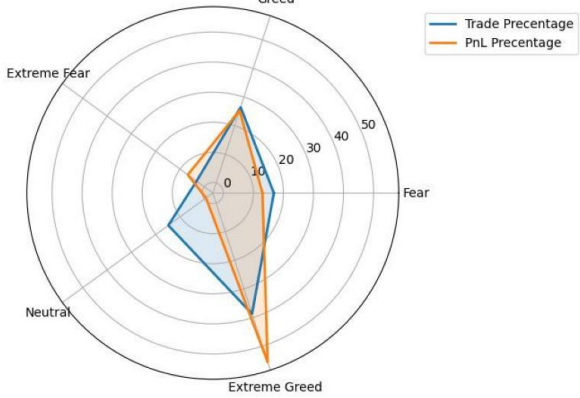
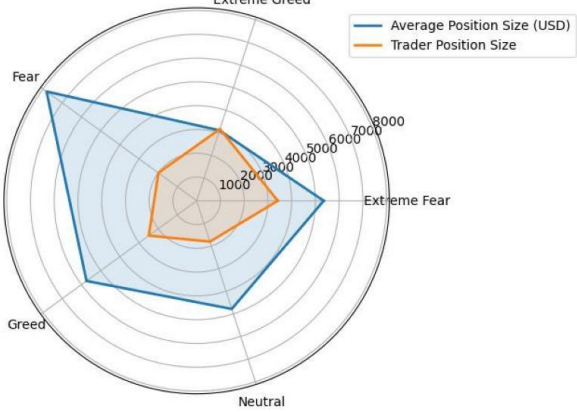
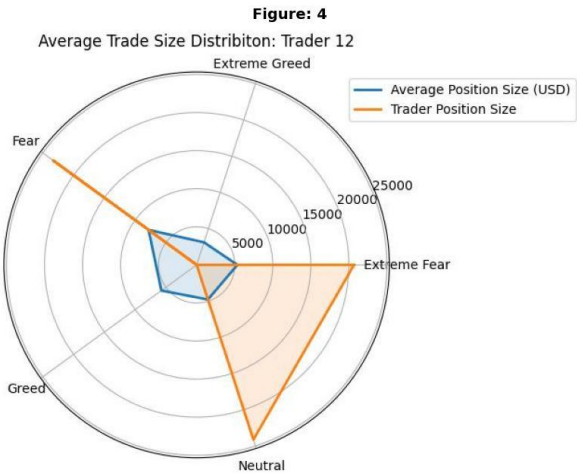
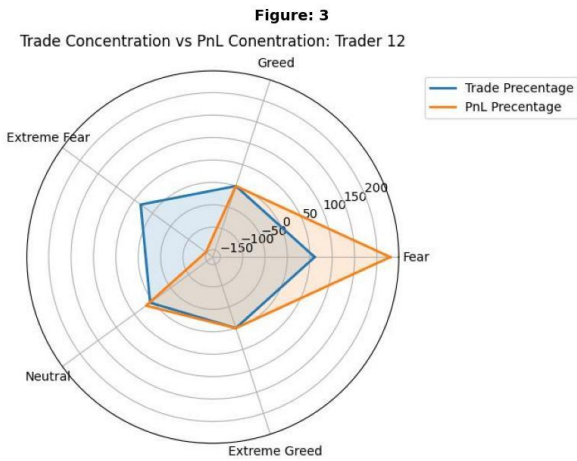
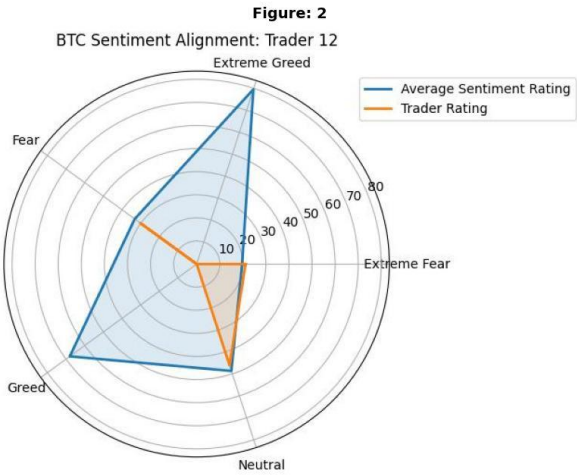
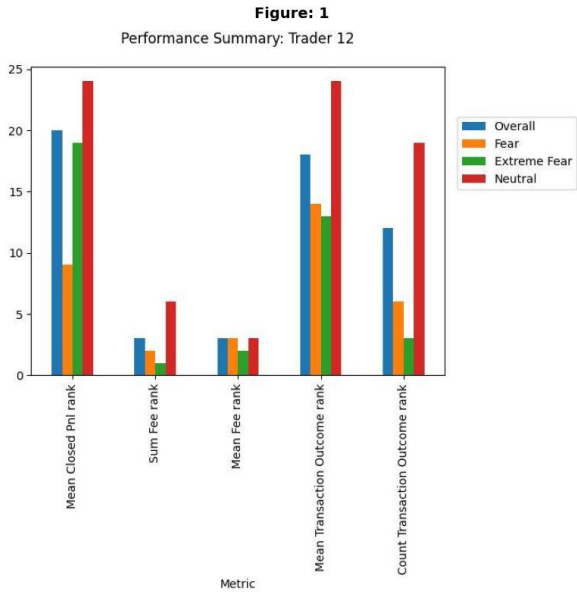


Figure 4
Average Trade Size Distribution: Trader 8



Combined BTC Market Sentiment Analysis for Trader 12
Account: 0xb899e522b5715391ae1d4f137653e7906c5e2115



Combined BTC Market Sentiment Analysis for Trader 2
Account: 0xbaaaf6571ab7d571043ff1e313a9609a10637864

Figure: 1
Performance Summary: Trader 2

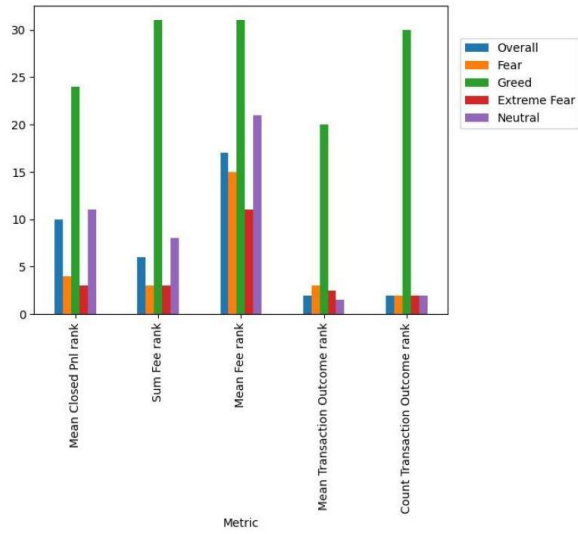


Figure: 2
BTC Sentiment Alignment: Trader 2

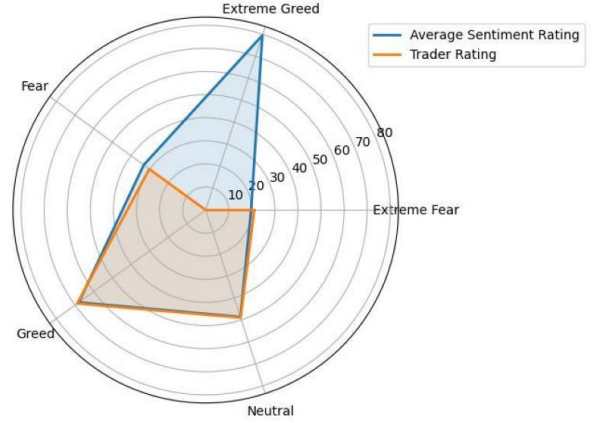
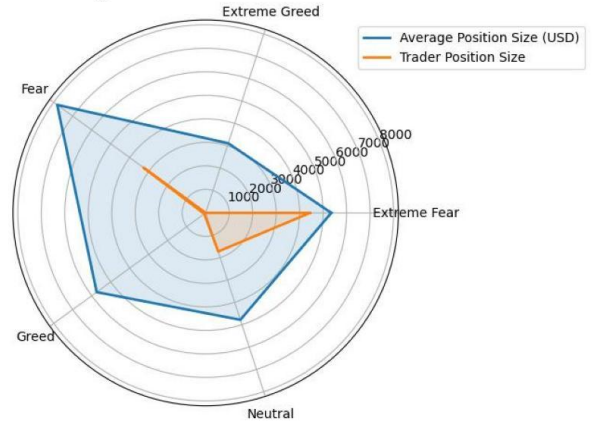


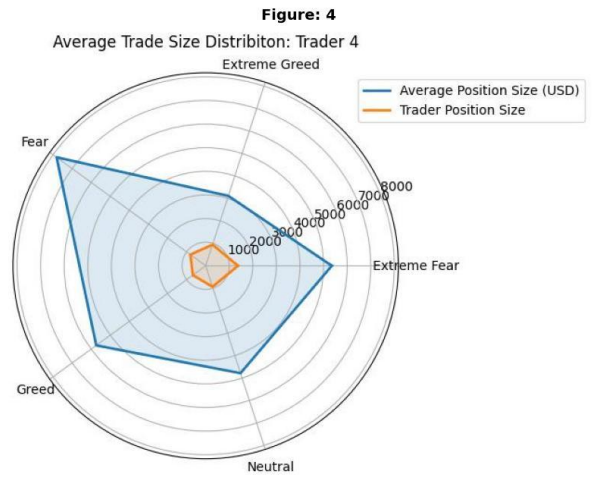
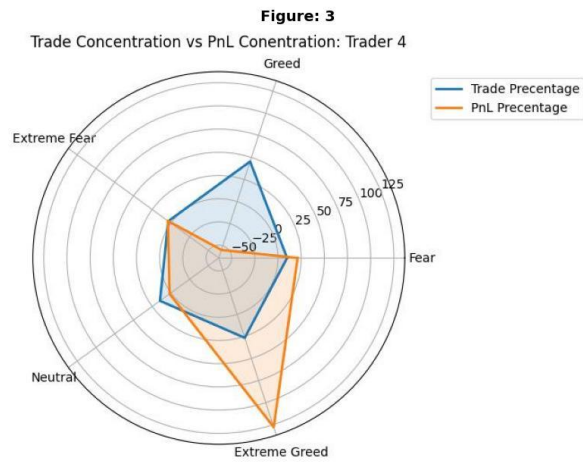
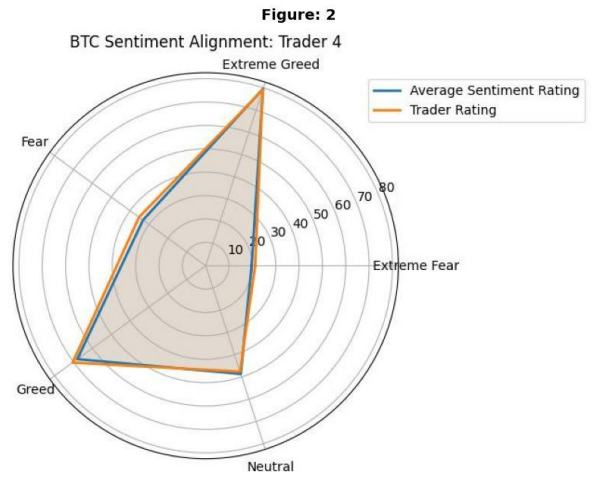
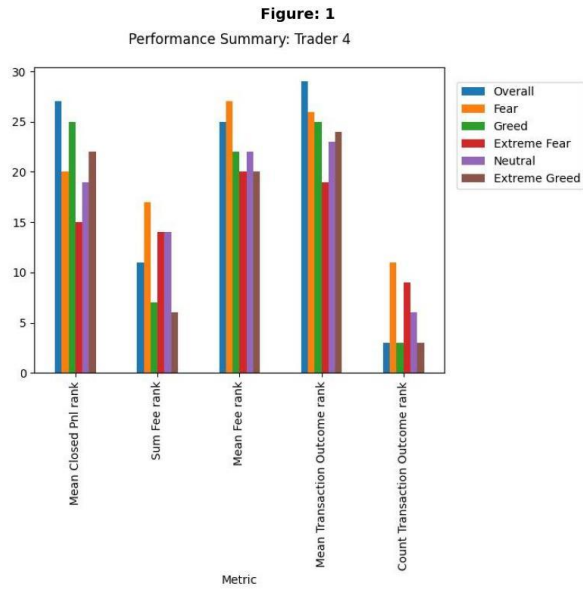
Figure: 3
Trade Concentration vs PnL Concentration: Trader 2



Figure: 4
Average Trade Size Distribution: Trader 2



Combined BTC Market Sentiment Analysis for Trader 4
Account: 0x8477e447846c758f5a675856001ea72298fd9cb5



Combined BTC Market Sentiment Analysis for Trader 5
Account: 0xb1231a4a2dd02f2276fa3c5e2a2f3436e6bfd23

Figure: 1
Performance Summary: Trader 5

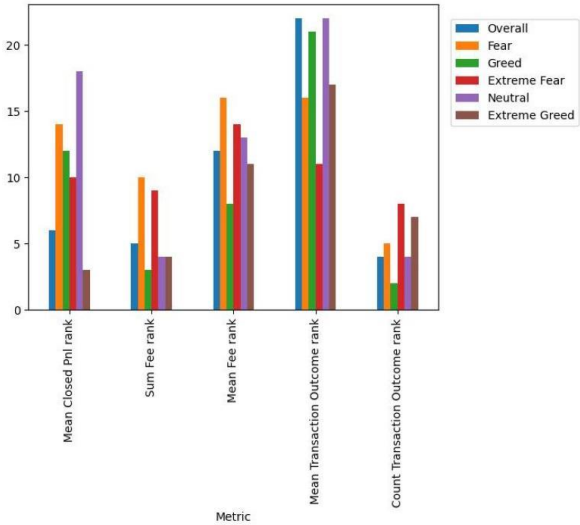


Figure: 2
BTC Sentiment Alignment: Trader 5

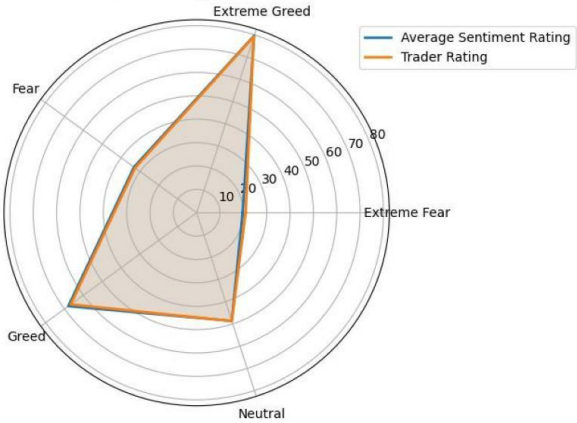
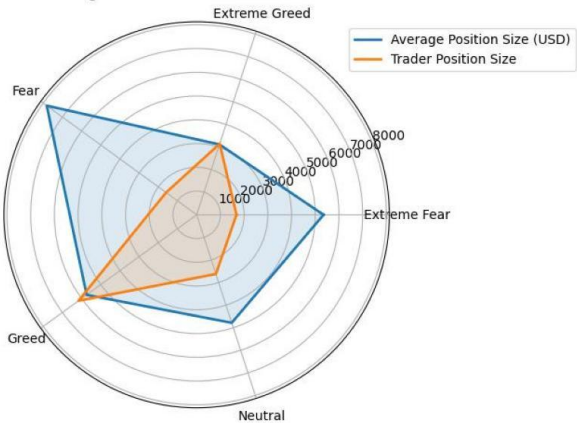


Figure: 3
Trade Concentration vs PnL Concentration: Trader 5



Figure: 4
Average Trade Size Distribution: Trader 5



Combined BTC Market Sentiment Analysis for Trader 9
Account: 0x47add9a56df66b524d5e2c1993a43cde53b6ed85

Figure: 1
Performance Summary: Trader 9

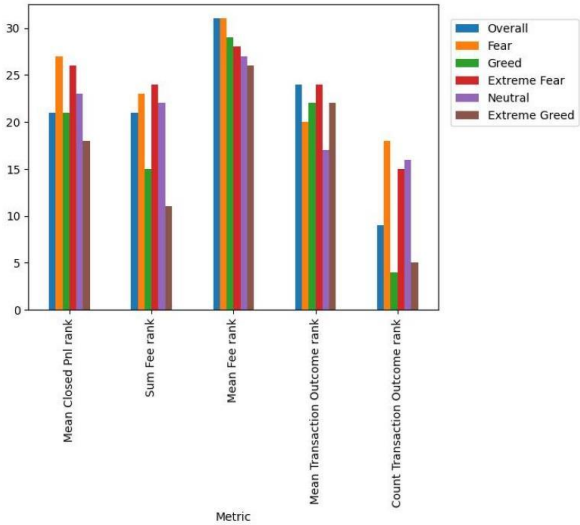


Figure: 2
BTC Sentiment Alignment: Trader 9

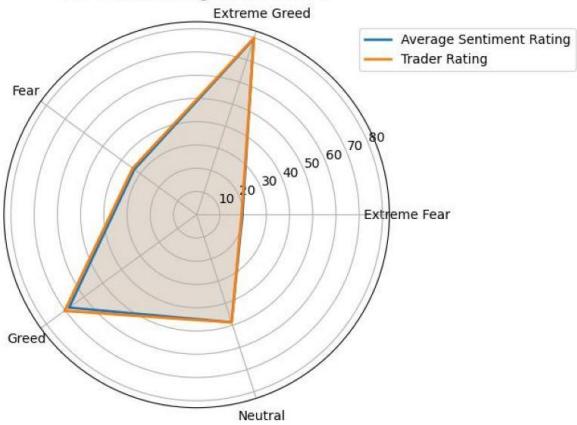
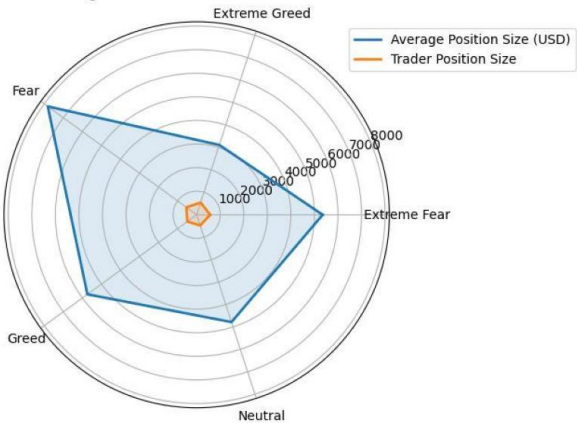


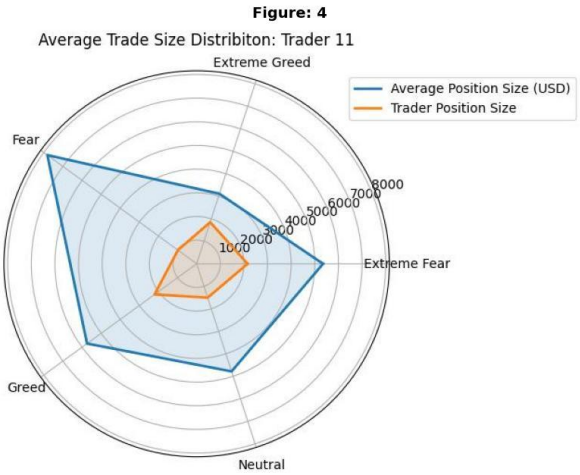
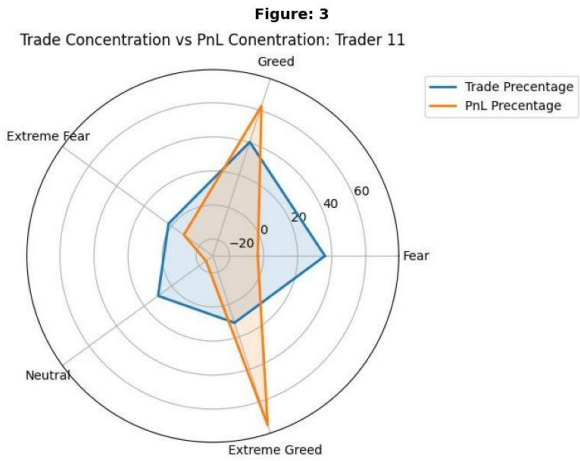
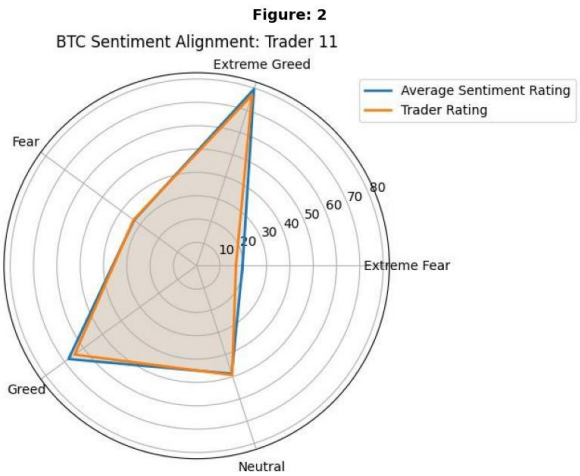
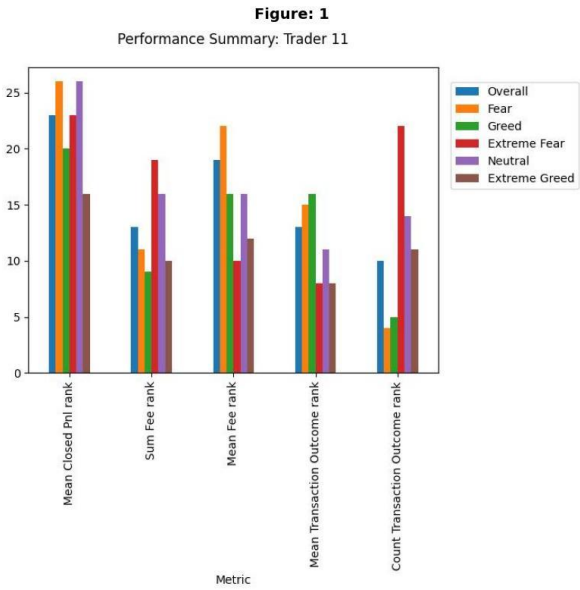
Figure: 3
Trade Concentration vs PnL Concentration: Trader 9



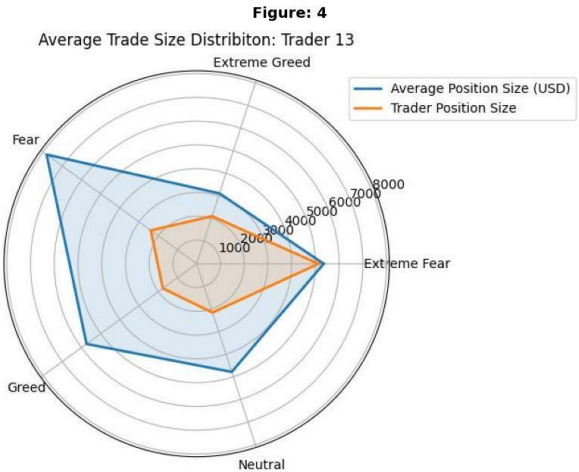
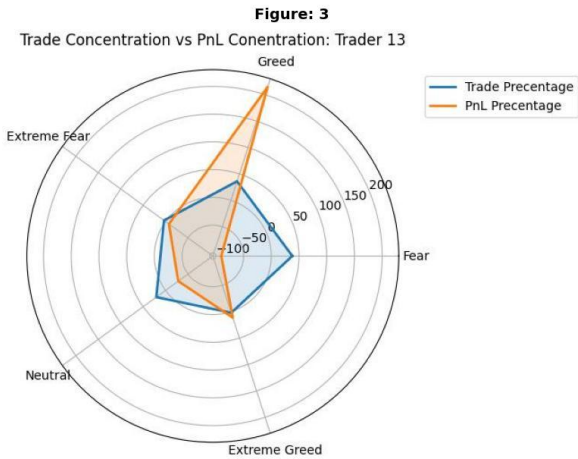
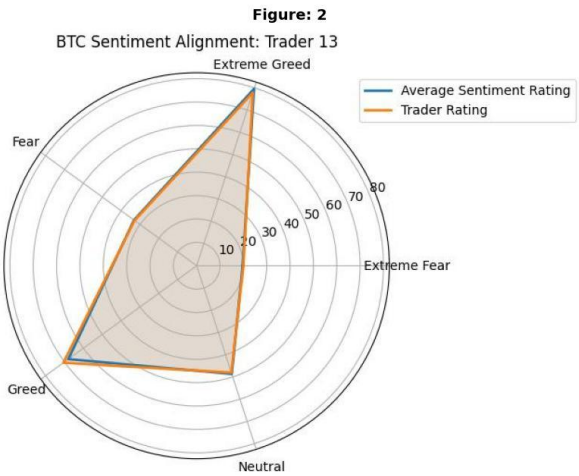
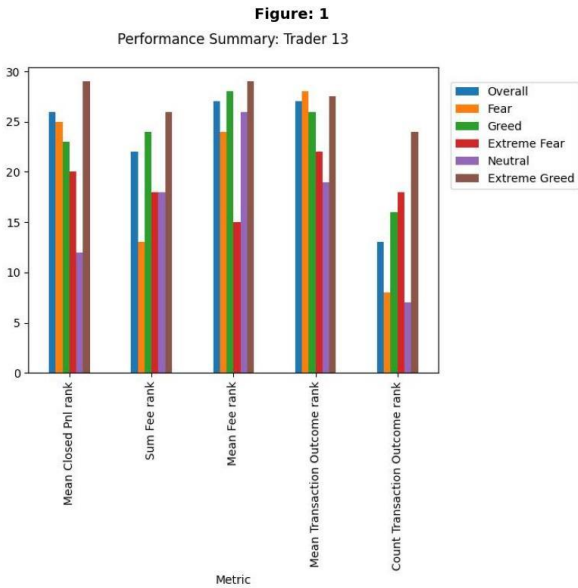
Figure: 4
Average Trade Size Distribution: Trader 9



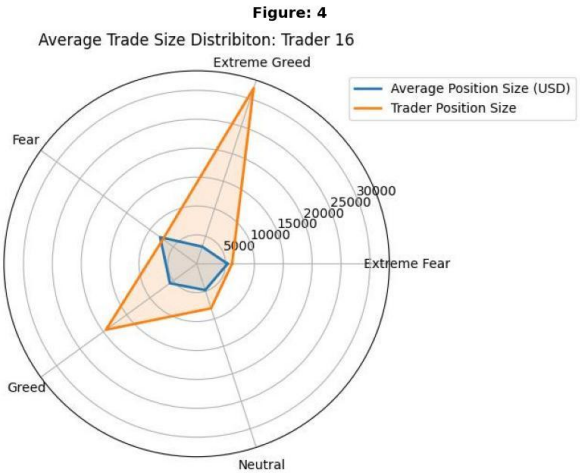
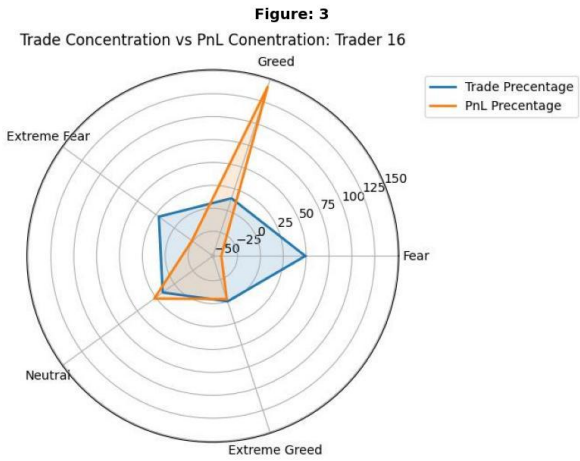
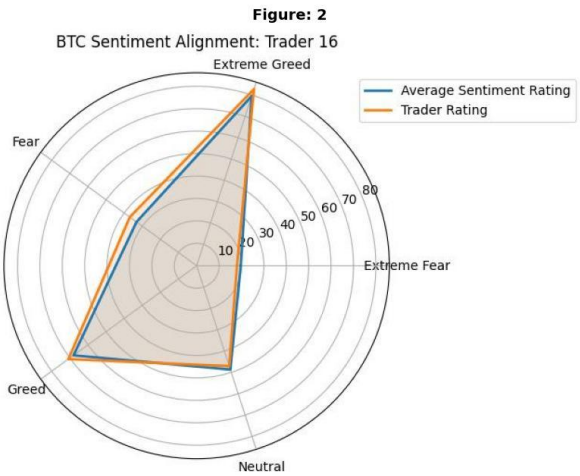
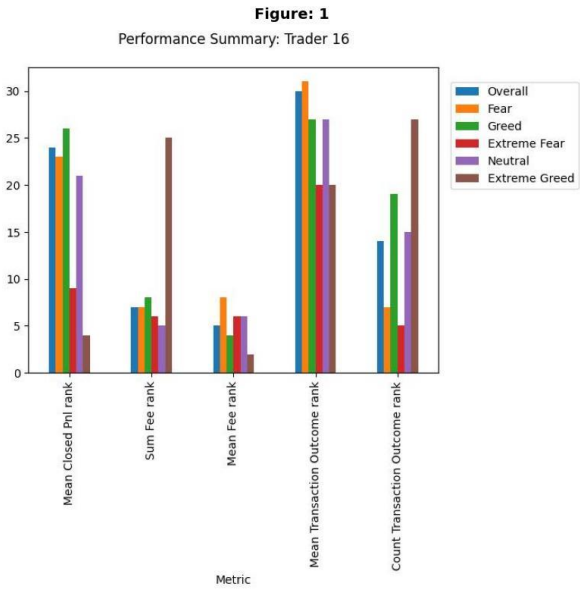
Combined BTC Market Sentiment Analysis for Trader 11
Account: 0x23e7a7f8d14b550961925fbfdaa92f5d195ba5bd



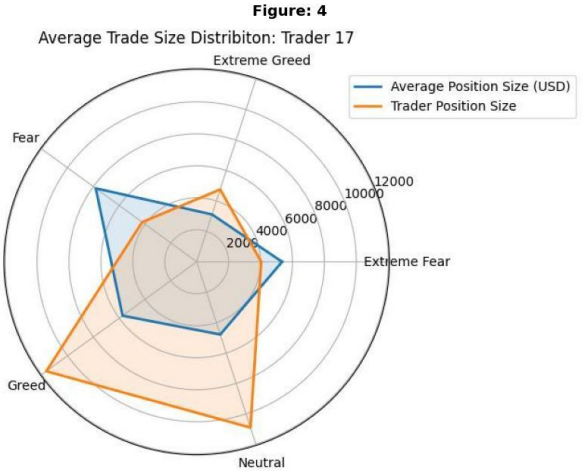
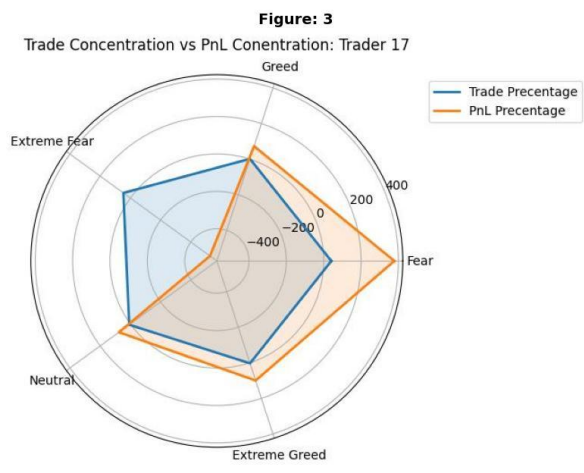
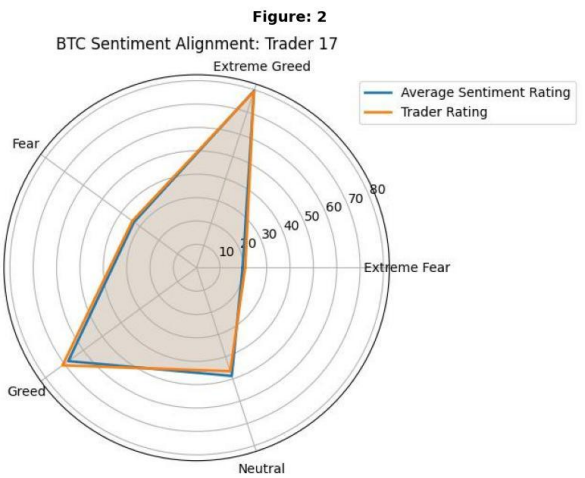
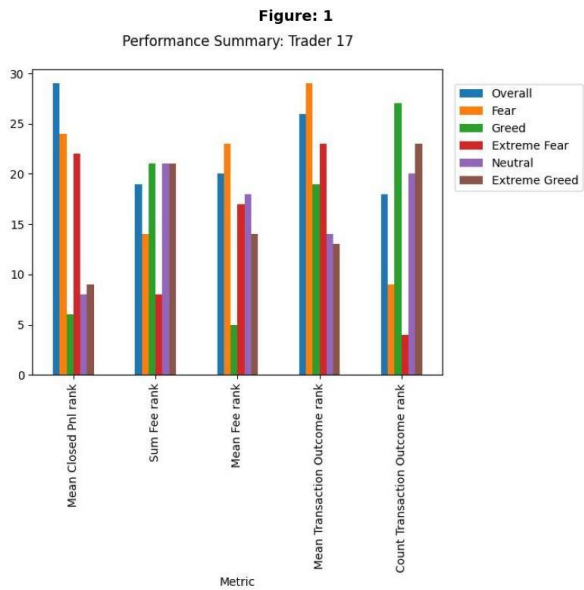
Combined BTC Market Sentiment Analysis for Trader 13
Account: 0x8170715b3b381dff7062c0298972d4727a0a63b



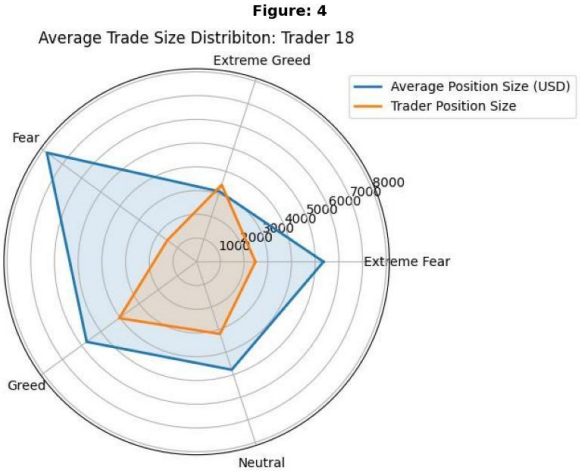
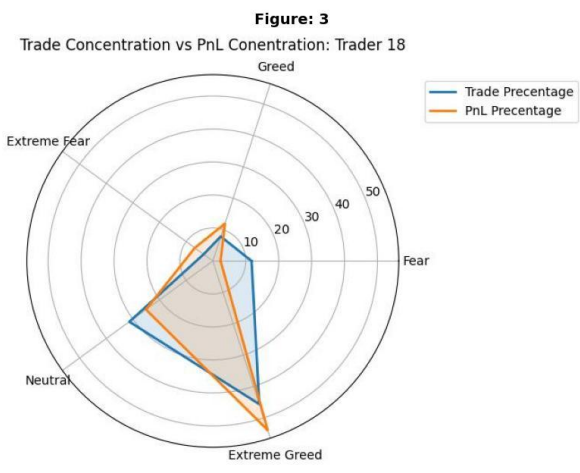
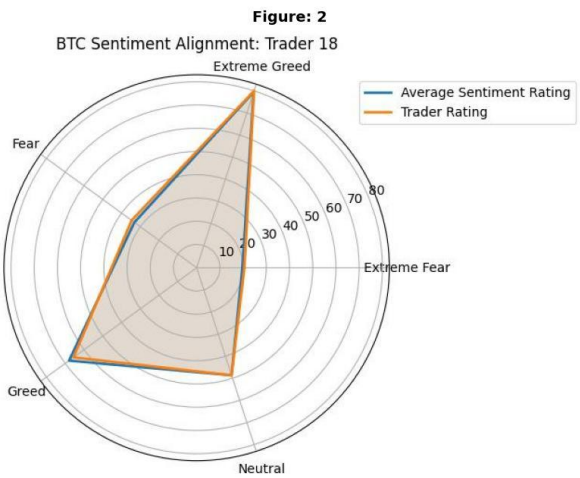
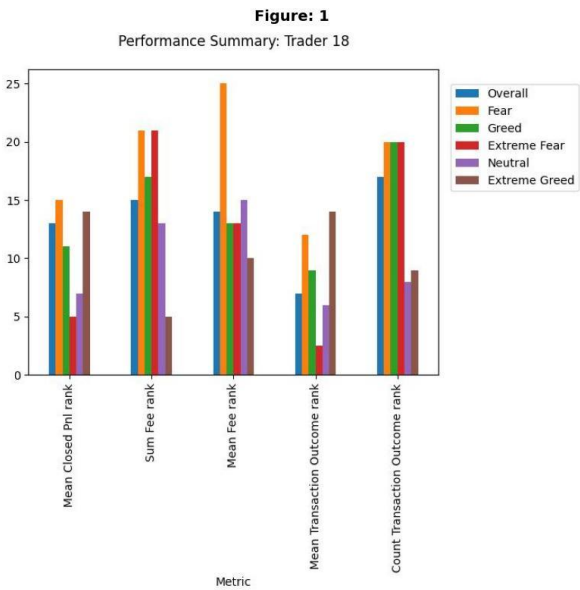
Combined BTC Market Sentiment Analysis for Trader 16
Account: 0x271b280974205ca63b716753467d5a371de622ab



Combined BTC Market Sentiment Analysis for Trader 17
Account: 0x39cef799f8b69da1995852eea189df24eb5cae3c



Combined BTC Market Sentiment Analysis for Trader 18
Account: 0x2c229d22b100a7beb69122eed721cee9b24011dd



Combined BTC Market Sentiment Analysis for Trader 19
Account: 0x92f17e8d81a944691c10e753af1b1baae1a2cd0d

Figure: 1
Performance Summary: Trader 19

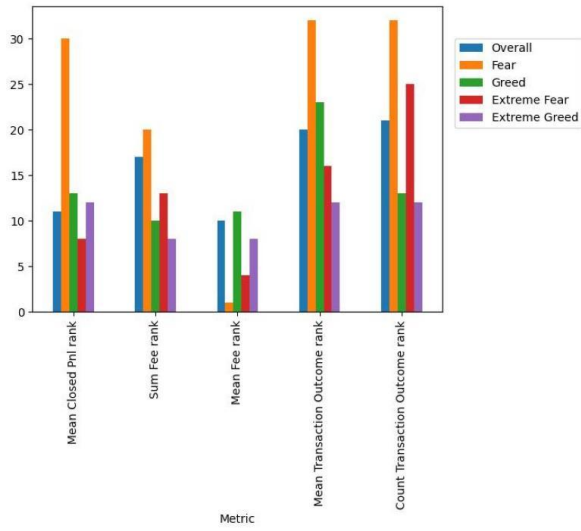


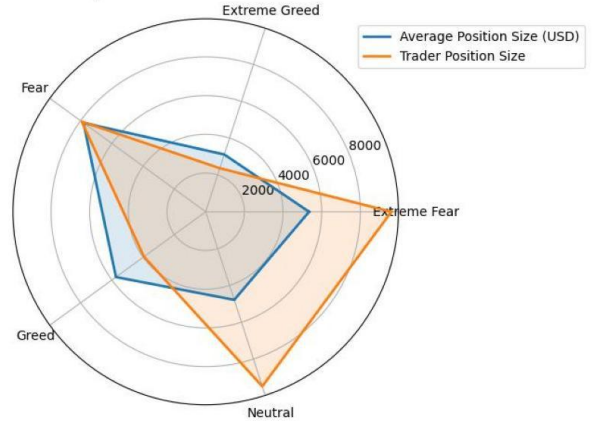
Figure: 2
BTC Sentiment Alignment: Trader 19



Figure: 3
Trade Concentration vs PnL Concentration: Trader 19



Figure: 4
Average Trade Size Distribution: Trader 19



Combined BTC Market Sentiment Analysis for Trader 20
Account: 0xbd5fead7180a9c139fa51a103cb6a2ce86ddb5c3

Figure: 1

Performance Summary: Trader 20

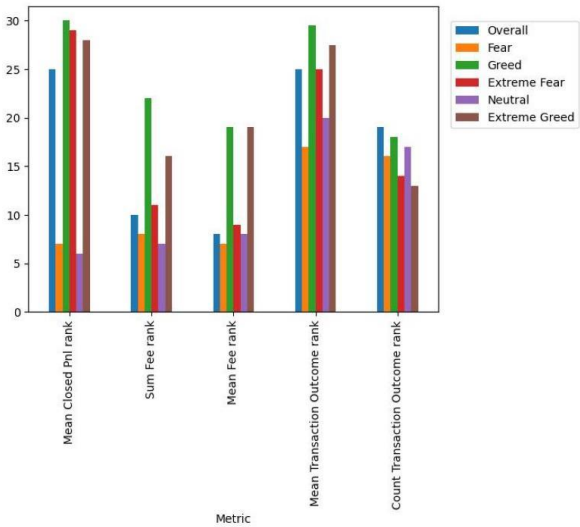


Figure: 2

BTC Sentiment Alignment: Trader 20

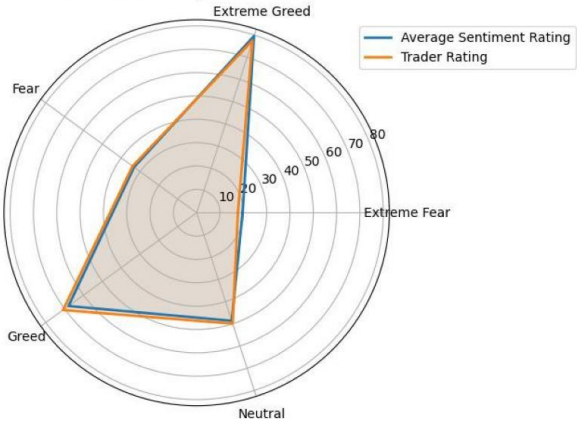


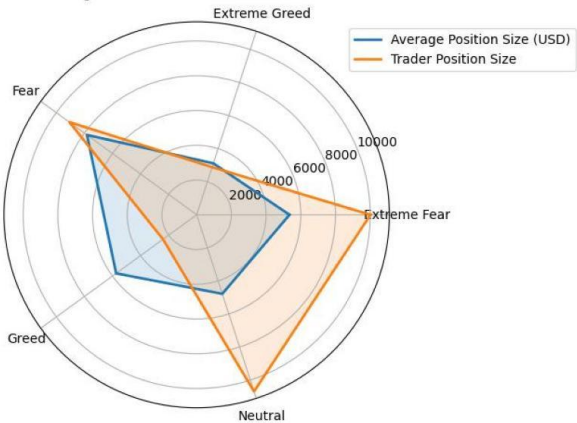
Figure: 3

Trade Concentration vs PnL Concentration: Trader 20

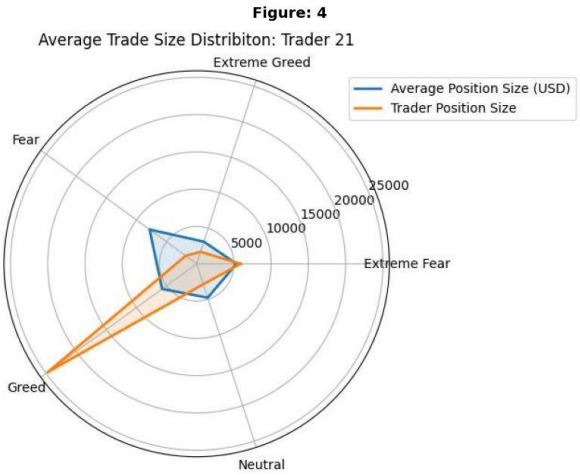
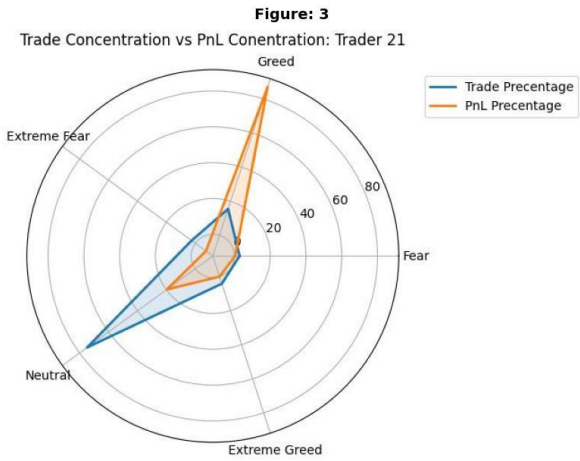
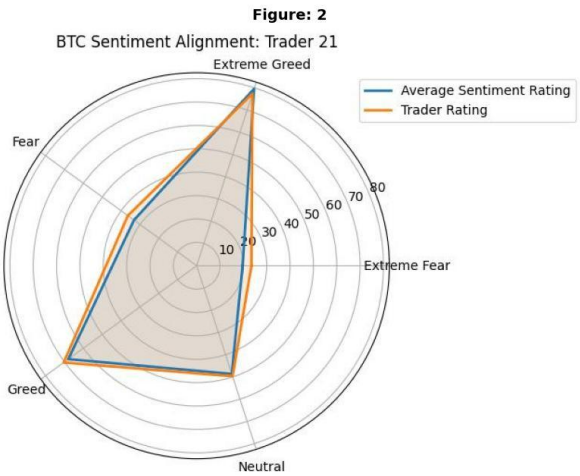
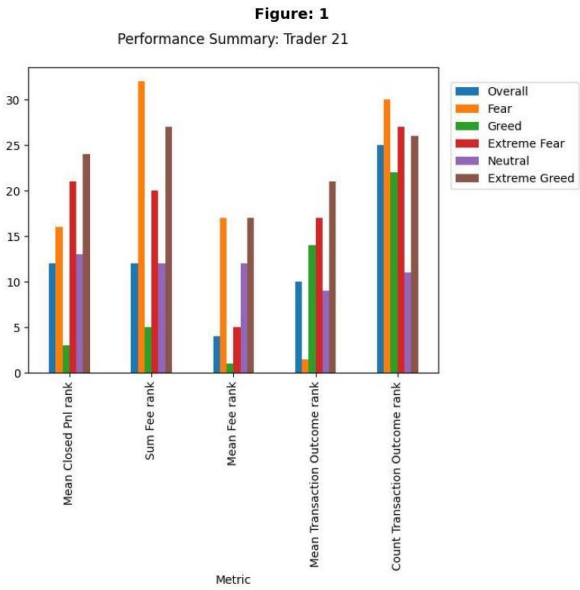


Figure: 4

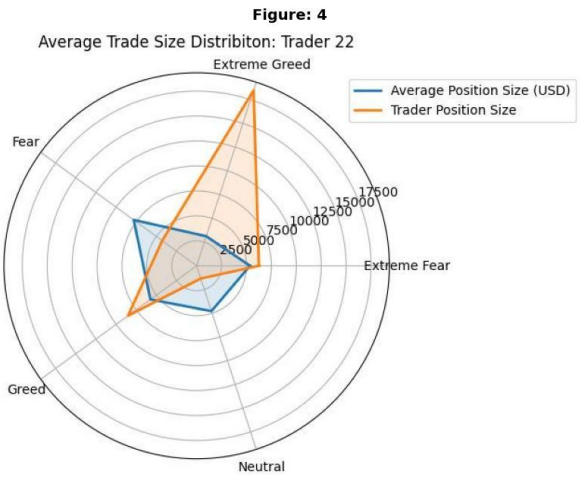
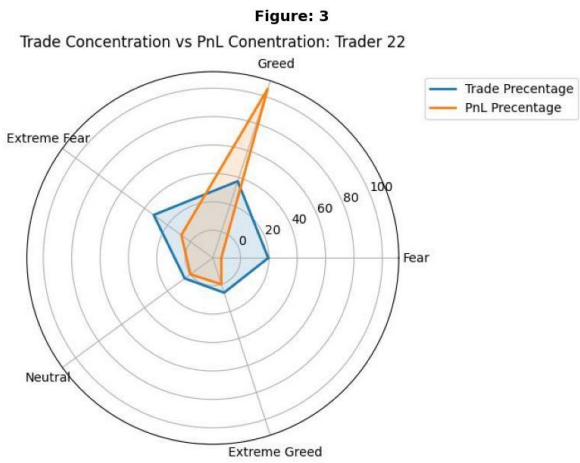
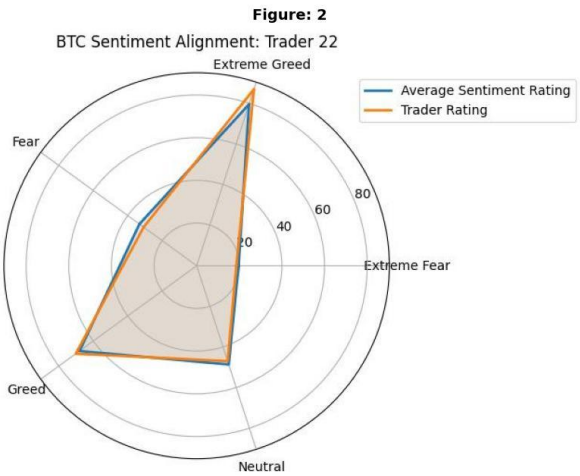
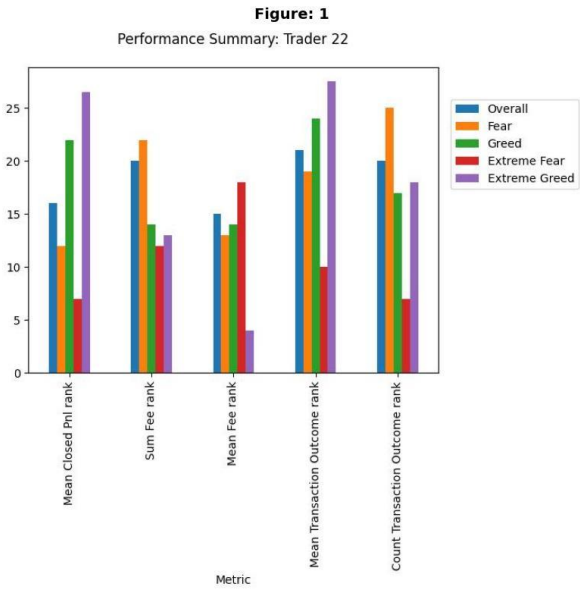
Average Trade Size Distribution: Trader 20



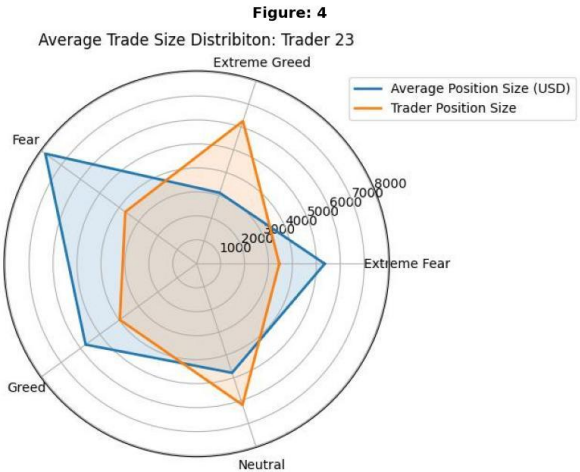
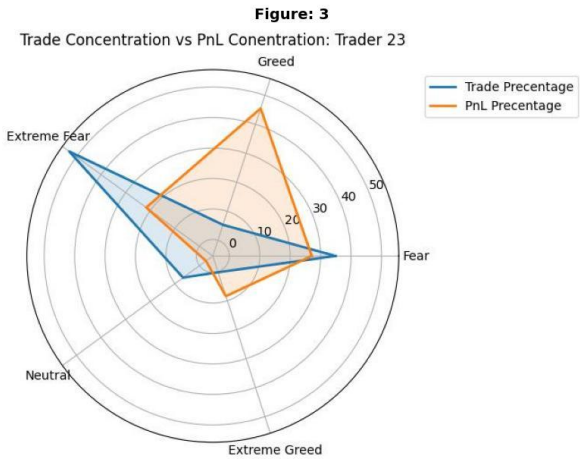
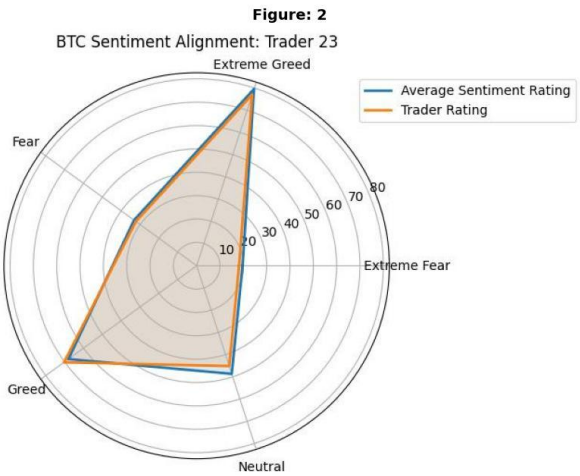
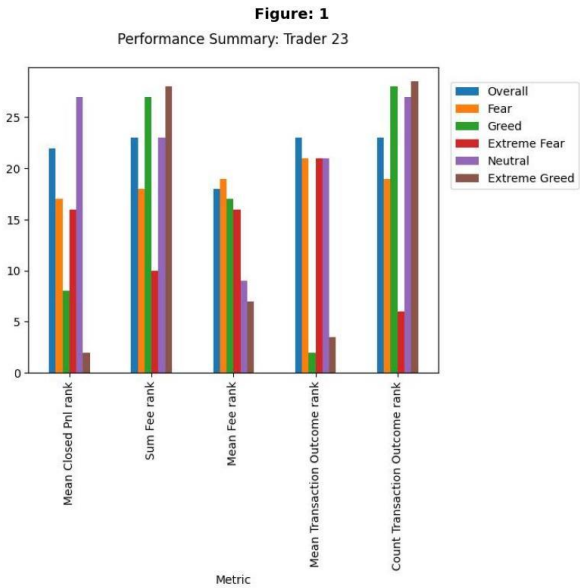
Combined BTC Market Sentiment Analysis for Trader 21
Account: 0x8381e6d82f1affd39a336e143e081ef7620a3b7f



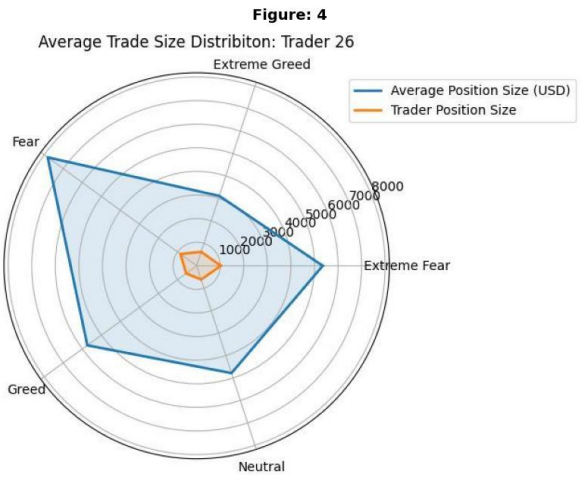
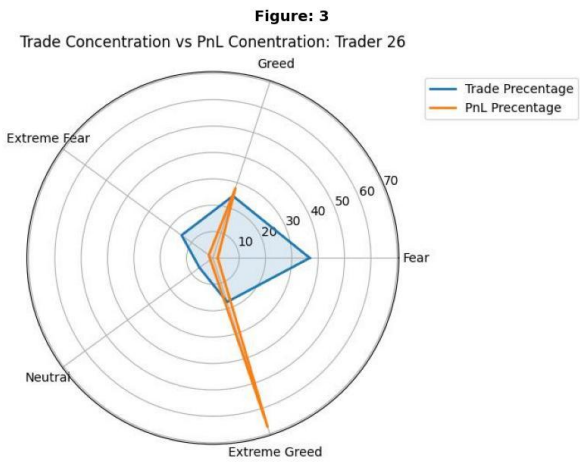
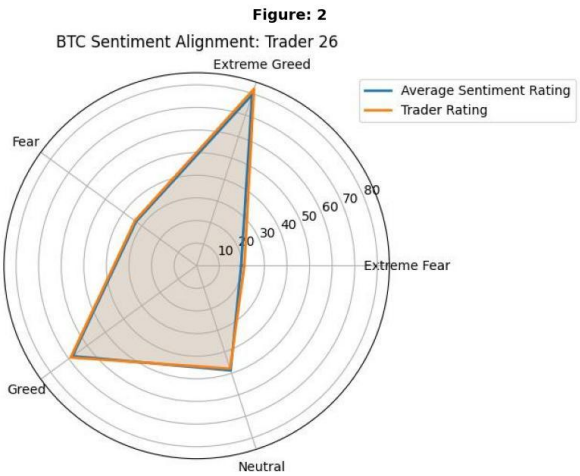
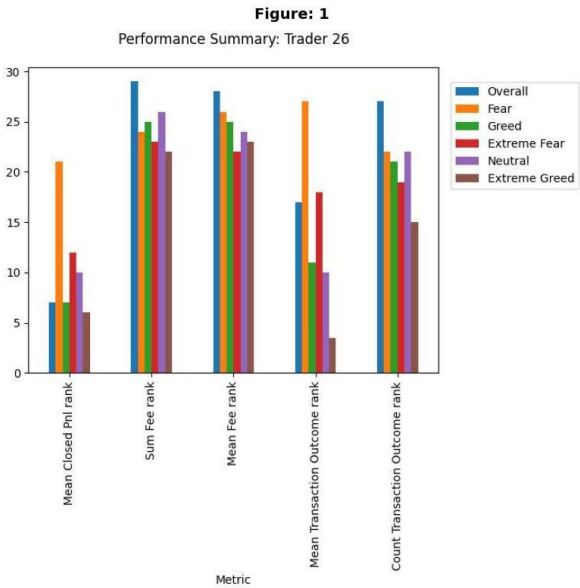
Combined BTC Market Sentiment Analysis for Trader 22
Account: 0x72743ae2822edd658c0c50608fd7c5c501b2afbd



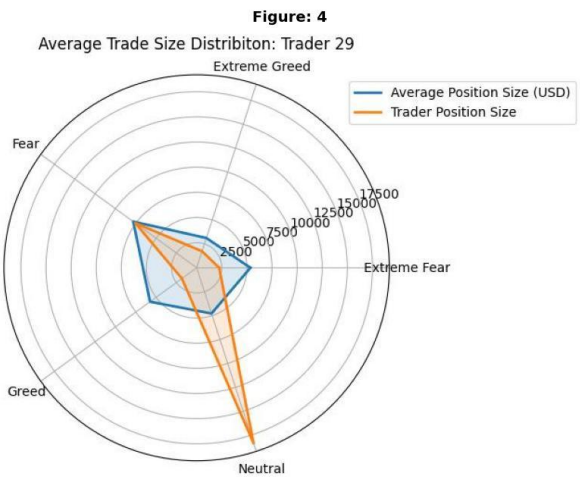
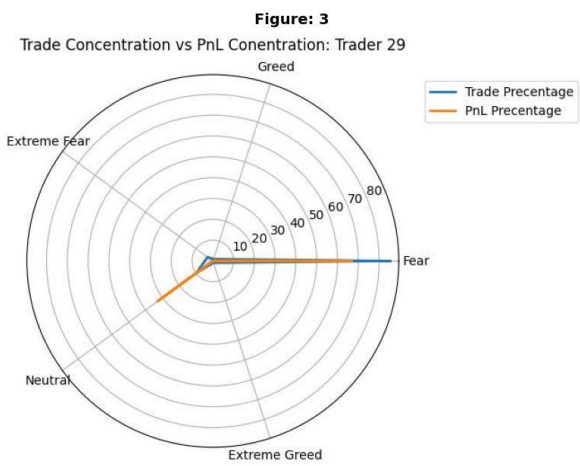
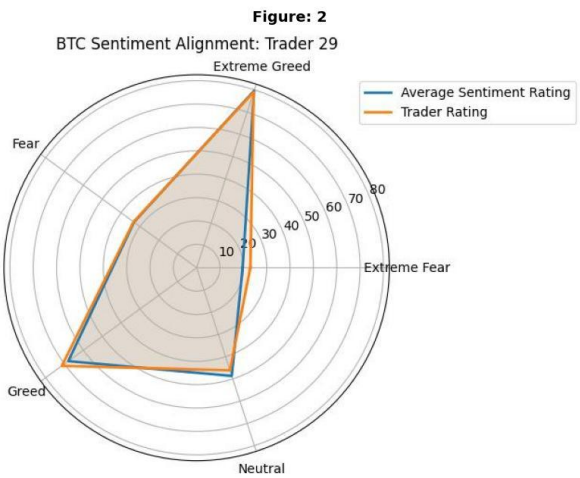
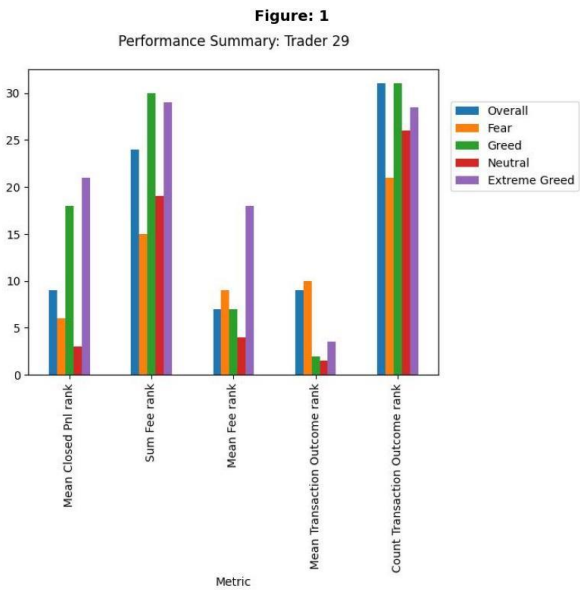
Combined BTC Market Sentiment Analysis for Trader 23
Account: 0x7f4f299f74eec87806a830e3caa9afa5f2b9db8f



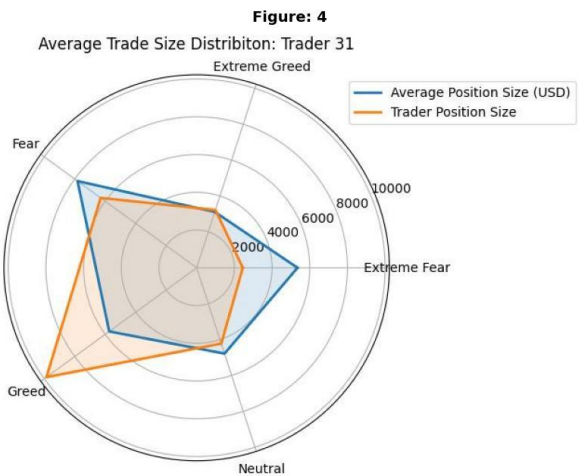
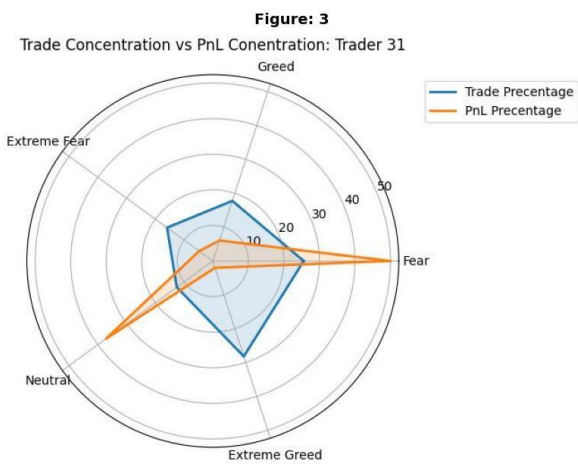
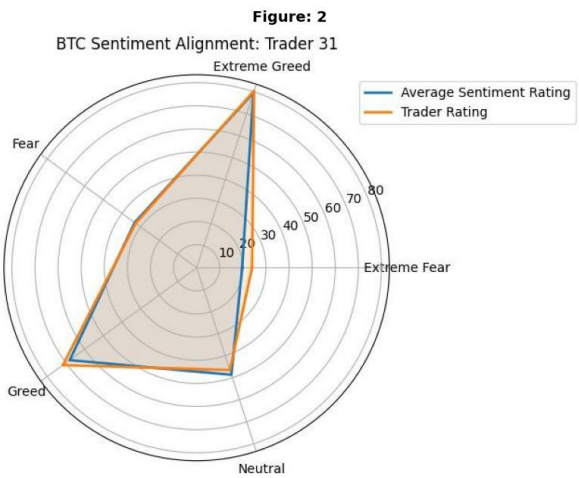
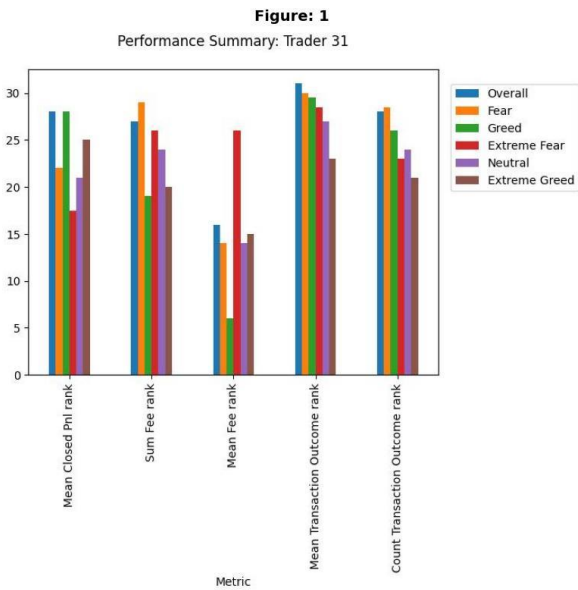
Combined BTC Market Sentiment Analysis for Trader 26
Account: 0x6d6a4b953f202f8df5bed40692e7fd865318264a



Combined BTC Market Sentiment Analysis for Trader 29
Account: 0xaf40fdc468c30116bd3307bcbf4a451a7ebf1deb



Combined BTC Market Sentiment Analysis for Trader 31
Account: 0x420ab45e0bd8863569a5efbb9c05d91f40624641



Combined BTC Market Sentiment Analysis for Trader 6
Account: 0x28736f43f1e871e6aa8b1148d38d4994275d72c4

Figure: 1
Performance Summary: Trader 6

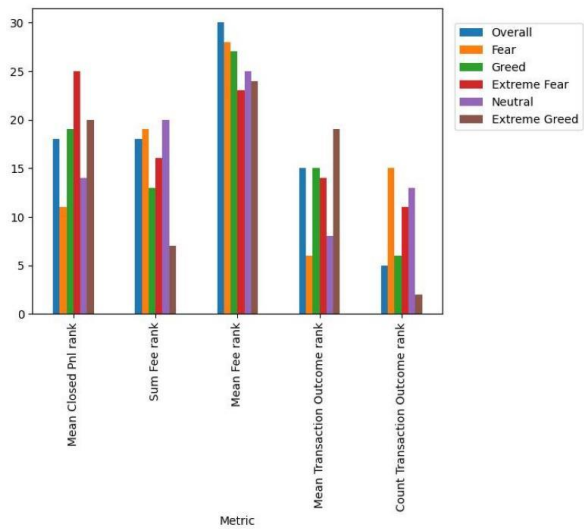


Figure: 2
BTC Sentiment Alignment: Trader 6

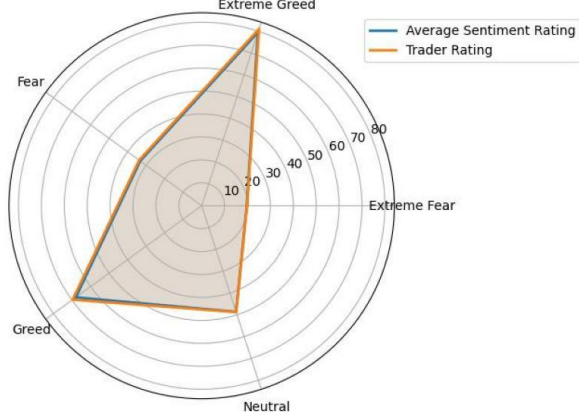
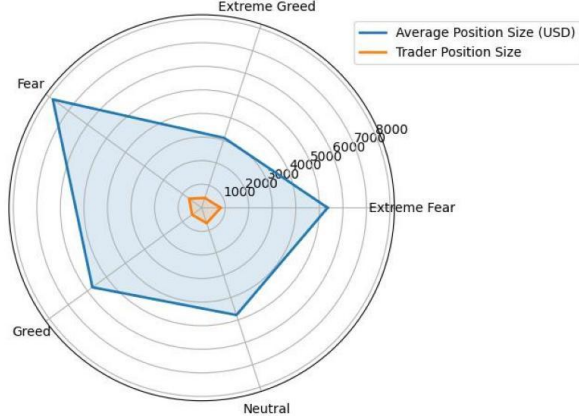


Figure: 3
Trade Concentration vs PnL Concentration: Trader 6

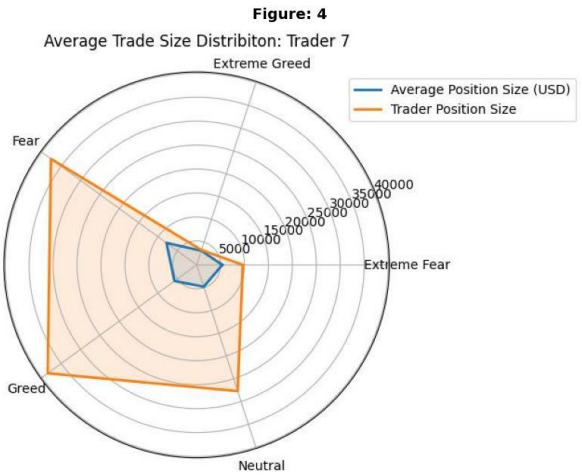
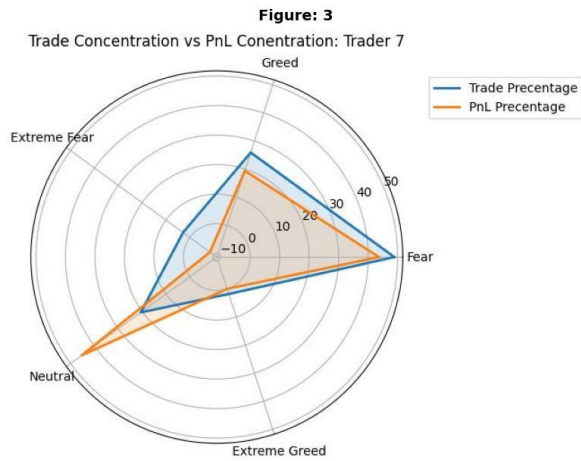
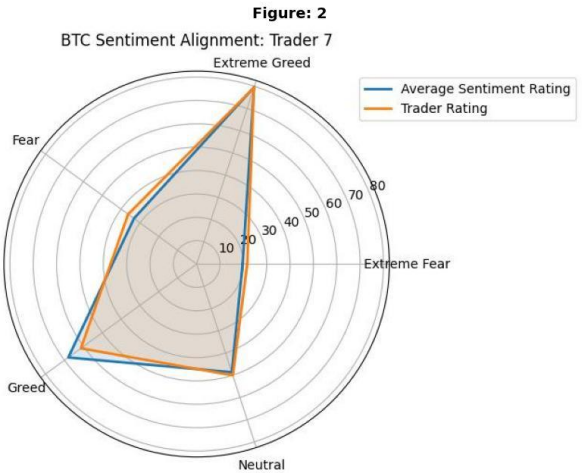
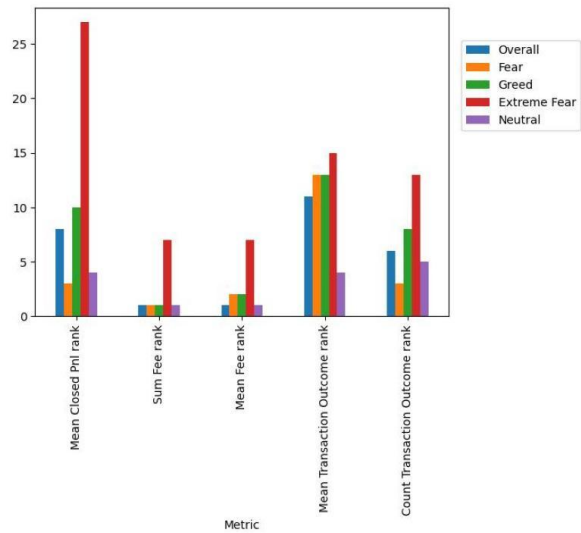


Figure: 4
Average Trade Size Distribution: Trader 6

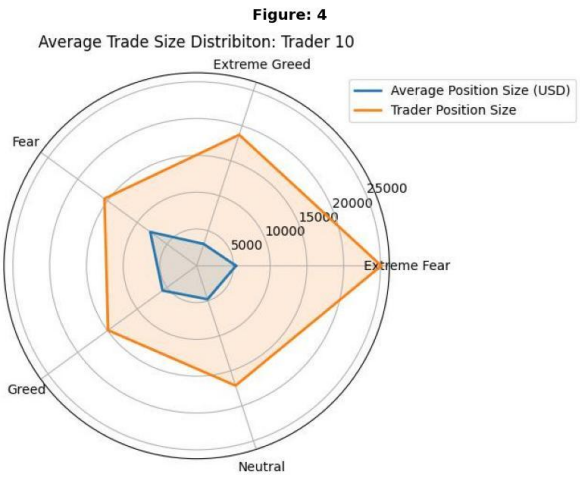
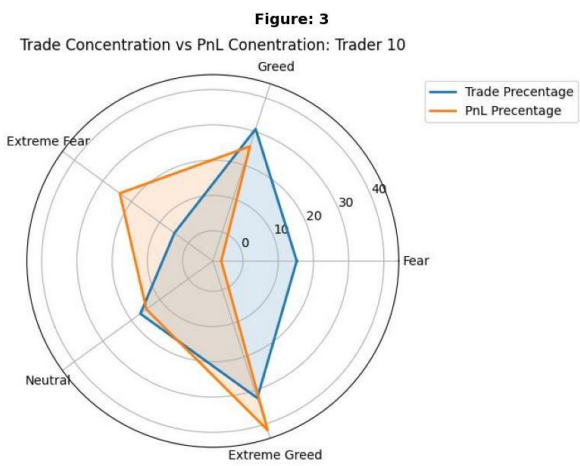
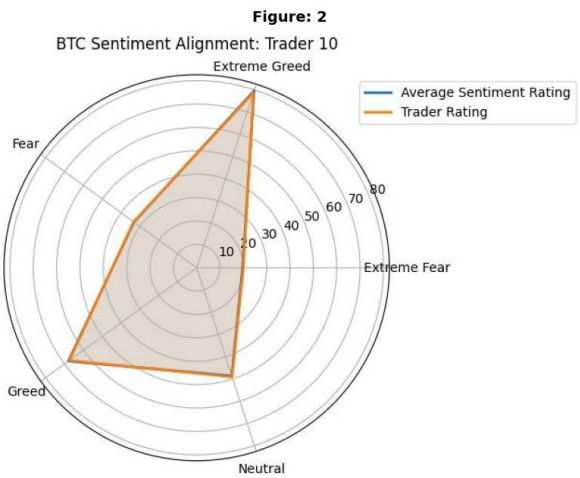
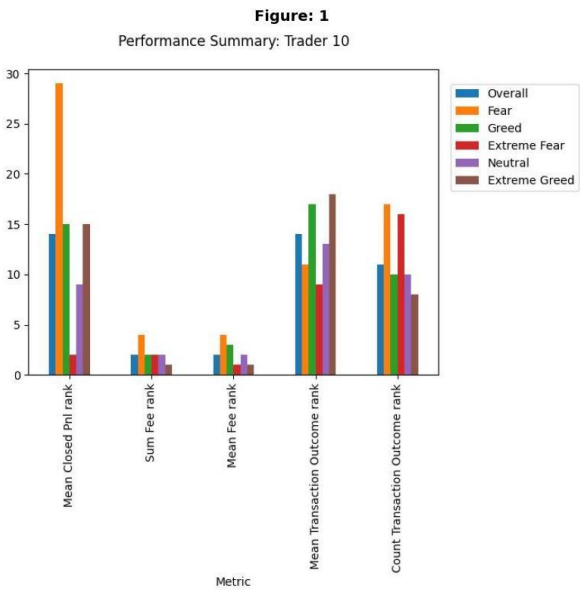


Combined BTC Market Sentiment Analysis for Trader 7
Account: 0x513b8629fe877bb581bf244e326a047b249c4ff1

Figure: 1
Performance Summary: Trader 7



Combined BTC Market Sentiment Analysis for Trader 10
Account: 0x4f93fead39b70a1824f981a54d4e55b278e9f760



Combined BTC Market Sentiment Analysis for Trader 25
Account: 0x430f09841d65beb3f27765503d0f850b8bce7713

Figure: 1
Performance Summary: Trader 25

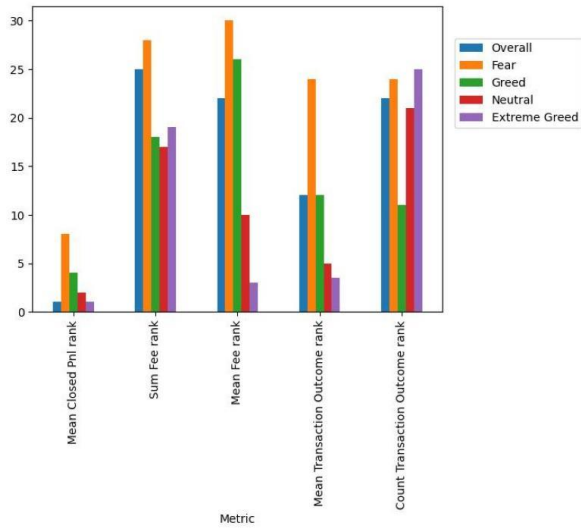


Figure: 2
BTC Sentiment Alignment: Trader 25

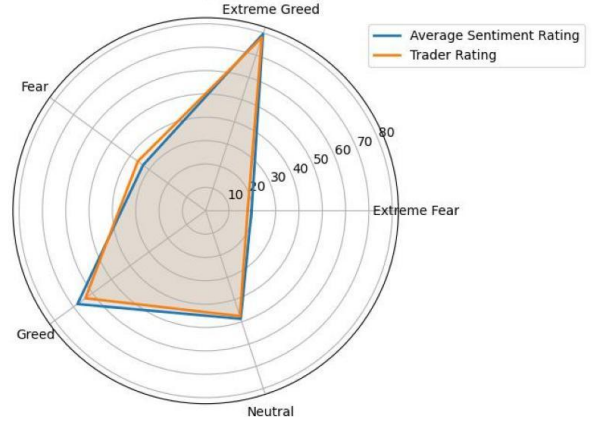


Figure: 3
Trade Concentration vs PnL Concentration: Trader 25

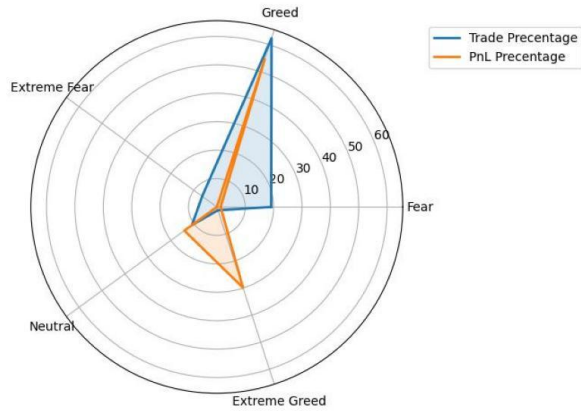
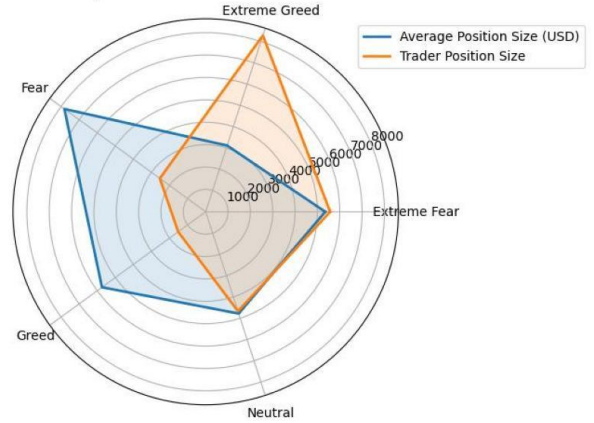
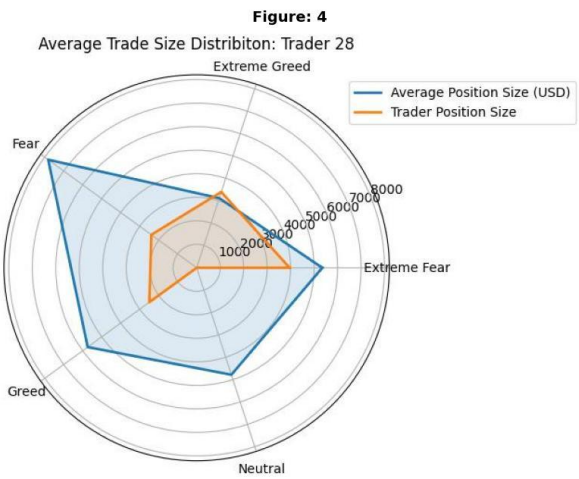
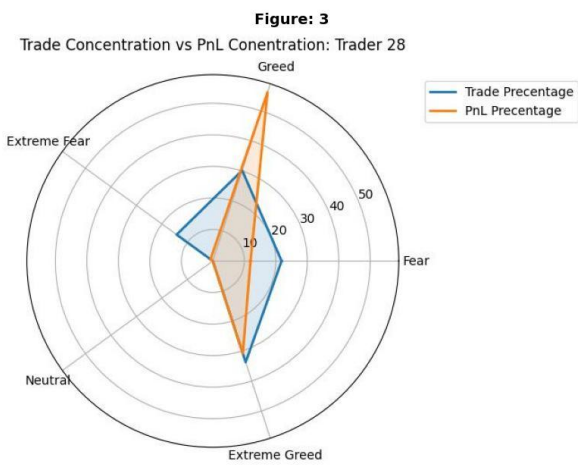
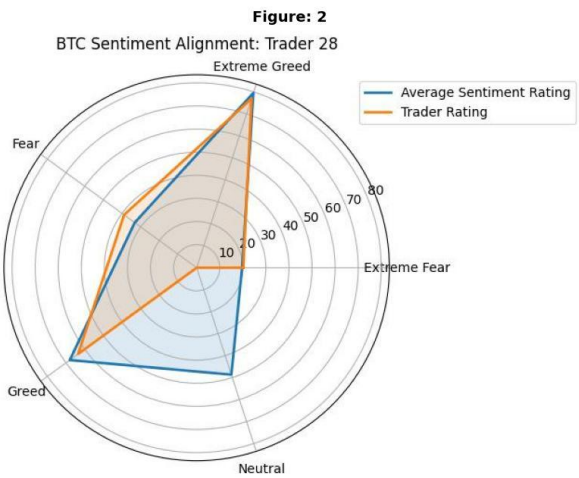
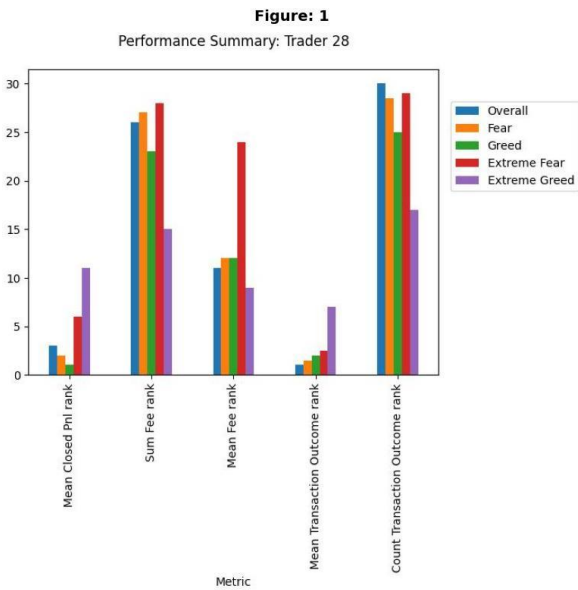


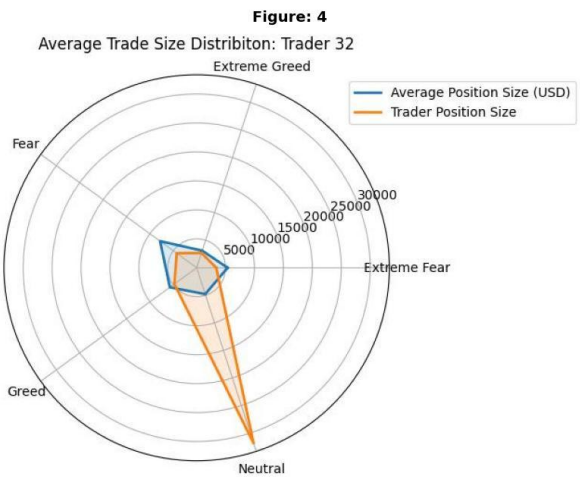
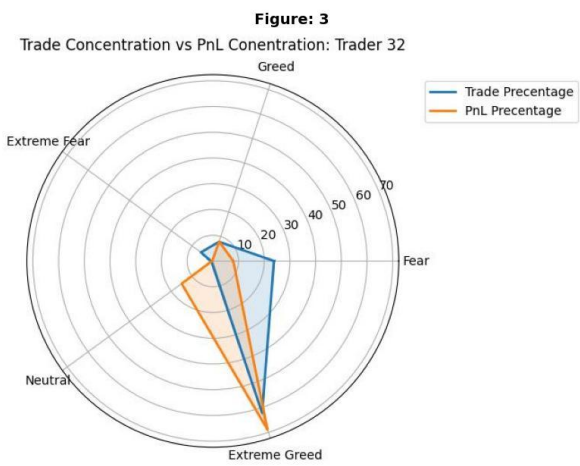
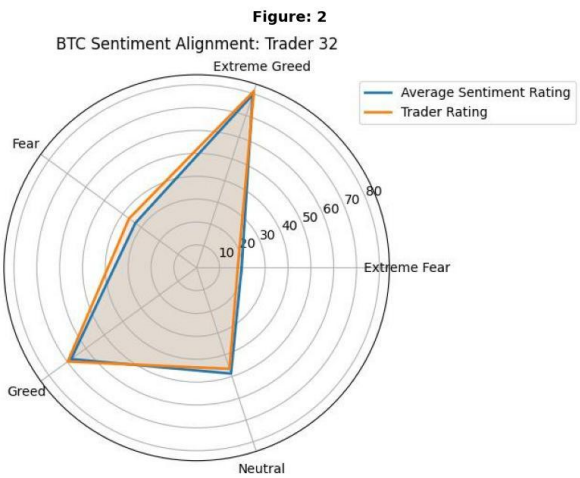
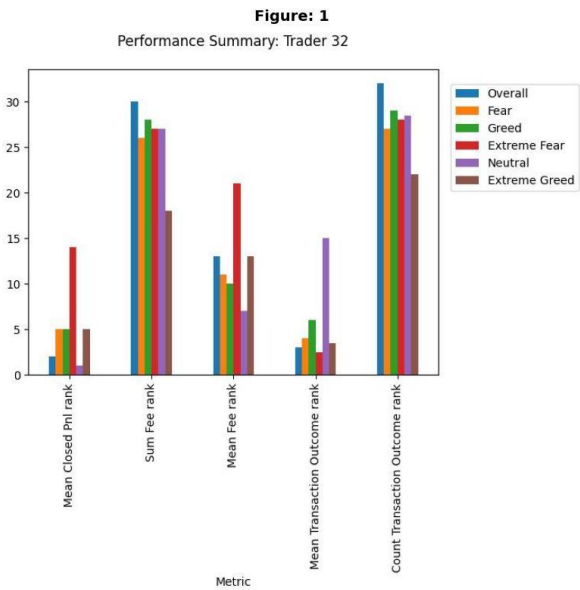
Figure: 4
Average Trade Size Distribution: Trader 25



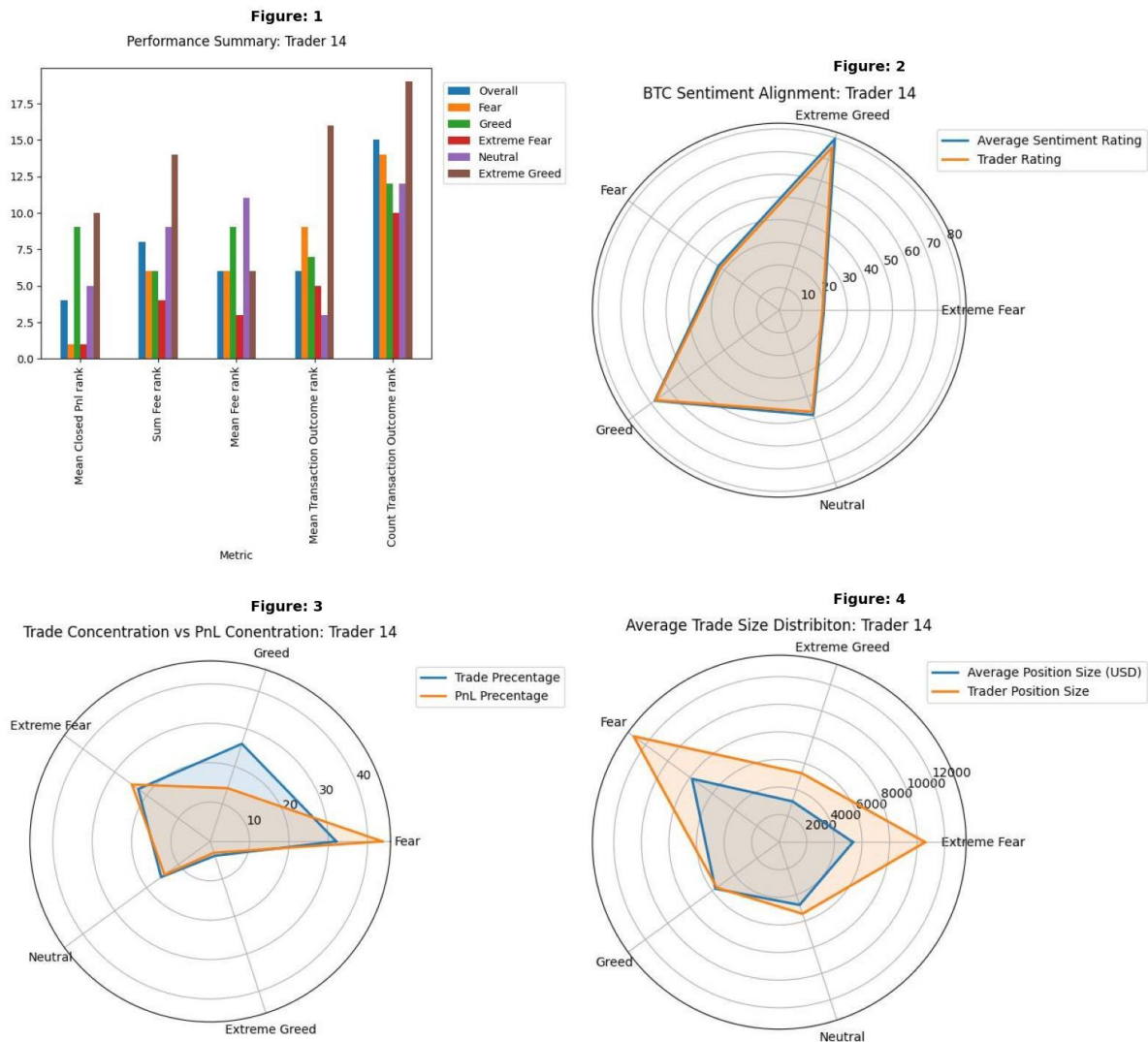
Combined BTC Market Sentiment Analysis for Trader 28
Account: 0xae5eacaf9c6b911fd53034a602c192a04e082ed



Combined BTC Market Sentiment Analysis for Trader 32
Account: 0x3f9a0adc7f04a7c9d75dc1b5a6ddd6e36486cf6



Combined BTC Market Sentiment Analysis for Trader 14 Account: 0x4acb90e786d897ecffb614dc822eb231b4ffb9f4



5. Areas of Improvement

1. More insights could have been drawn if an explanation of what the data in each column meant for the `historical_data.csv` file was available.
2. The ambiguity with the difference between ["Close Long", "Open Short"] and "SELL" as well as that between ["Open Long", "Close Short"] and "BUY" in the Direction column of `historical_data.csv` caused the analysis to of total and average pnl to be limited to ["Close Long", "Open Short", ["Open Long", "Close Short"].
3. Analysis of trader behavioral traits by Direction i.e. ["Close Long", "Open Short", ["Open Long", "Close Short"] in different regimes of market sentiment could have been incorporated.

6. Additional Notes

An additional function `get_account_details( )` has been defined for detailed analysis of trader performance. This function allows the user to visualize the coinwise performance of traders segregated by market sentiment regimes, as well as the individual trades executed by the trader during the time period.